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Project Overview, Database Setup + Collection Creation and Importation of Data

Project Title:

Scientific Research Project Database in MongoDB

Objective:

To design and implement a document-based database using MongoDB that stores comprehensive details about scientific research projects across various disciplines and institutions in Ireland, Scotland, Wales, and England. This project focuses on advanced data modeling and retrieval techniques using MongoDB Shell and Compass.

Scope:

- Store details of scientific projects including description, subject area, and project value
- Capture nested data structures such as universities and project leaders
- Include arrays for sponsors and resources used in each project
- Cover projects conducted at institutions in Ireland, Scotland, Wales, and England
- Focus on retrieval queries using MongoDB Shell and aggregation pipelines
- Build a schema flexible enough for complex data types while ensuring query efficiency

Document Structure in *projects* Collection:

Each document in the `projects` collection contains the following keys:

1. Top-Level Fields:

- `_id` (int)
- `Description` (string)
- `Start_Year`, `End_Year` (int or string)
- `Discipline` (string)
- `Subject_Area` (string)
- `Num_Personnel` (int)
- `Project_Value` (int)
- `Progress_Report` (string)

2. Arrays:

- **Sponsor** (array of strings)
 - Example values: "EU", "IRC", "WHO", "Private Donation", "Department of Education", "Government of Ireland"
- **Resources** (array of strings)
 - Example values: "High-Performance Computers", "Laptops", "Quantum Lab", "Cloud Computing Credits", "3D Printers", "AI Workstations", "Microscopes", "Spectrometers"

3. Embedded Subdocuments:

- **University:**
 - Fields: Name, City, Num_Undergrad, Specialism, Annual_Budget
- **Proj_Lead** (Project Leader):
 - Fields: Name, Qualification, Salary, Specialism, Num_Publication

Relationships (Implied by Embedded Documents):

1. Each project is associated with **one university** and **one project lead**
2. No normalization is enforced (denormalized structure typical of NoSQL)
3. The structure supports quick access to all related information within a single document

Expected Outcomes:

- A MongoDB collection (`projects`) that can handle nested and array data effectively
- Capability to retrieve, filter, and summarize project data using MongoDB Shell
- Well-designed queries to:
 - Search projects based on multi-criteria filters
 - Generate statistics (e.g. project count per sponsor, total project value per university)
- Demonstration of MongoDB proficiency through CRUD reads and aggregation pipelines

Database Setup

Database Setup and Collection Creation

The MongoDB environment was installed locally on Windows. The following steps outline the setup process:

Dataset Creation and Import Process:

1. Dataset Construction

A custom dataset of 1,000 research project records was generated, each representing a unique scientific research initiative across multiple disciplines and UK/Irish universities. The records include diverse fields such as project description, timeframes, discipline, personnel, and nested university and project leader information.

2. Conversion to JSON Format

The dataset was cleaned and exported into a JSON array file (*Msc2025_Scientific_Research_Project_A_Karim.json*), ensuring it followed proper JSON formatting (e.g., corrected quote marks, removed tab characters, validated brackets).

3. MongoDB Tools Used

- **mongod**: This daemon was started to launch the MongoDB server, enabling local database operations:

"C:\Program Files\MongoDB\Server\8.0\bin\mongod.exe"

(Note: This path may vary depending on the installation directory.)

- **mongosh**: The interactive shell was used to interact with MongoDB:
- **mongoimport**: Using the mongoimport tool, the JSON file was successfully imported into a MongoDB collection called projects within a database named researchDB. The import command used was:

```
mongoimport --db researchDB --collection projects --file  
"Msc2025_Scientific_Research_Project_A_Karim.json" --jsonArray
```

The MongoDB server was running locally on the default port (27017) during the import.

4. Using the Database in Shell

Once in the shell, the following command ensures queries are run within the intended database:

```
test> use researchDB  
switched to db researchDB
```

Database Setup Overview:

- **Database Name:** researchDB
- **Collection Name:** projects

Each document is a self-contained project entry, with nested subdocuments and array fields.

Array Fields in the Collection:

Sponsor (Array of Strings)

Funding bodies that support each project:

- "EU"
- "IRC"
- "WHO"
- "Private Donation"
- "Department of Education"
- "Government of Ireland"

Resources (Array of Strings)

Tools, infrastructure, and computing resources used:

- "High-Performance Computers"
- "Laptops"
- "Quantum Lab"
- "Cloud Computing Credits"
- "3D Printers"
- "AI Workstations"
- "Microscopes"
- "Spectrometers"

Embedded Documents:

University Example:

```
...
[
  {
    University: {
      Name: 'Trinity College Dublin',
      City: 'Dublin',
      Num_Undergrad: 13000,
      Specialism: 'Astrophysics',
      Annual_Budget: 32321
    }
  }
]
researchDB> |
```

Proj_Lead Example:

```
[
  {
    Proj_Lead: {
      Name: 'Dr. Aisha Rahman',
      Qualification: 'PhD',
      Salary: '78000',
      Specialism: 'Climate Change',
      Num_Publication: 28
    }
  }
]
```

Sample Document:

```
[
  {
    _id: 3,
    Description: 'Leveraging Quantum Computing for Understanding Protein Folding Mechanisms in Biological Systems',
    Start_Year: '2018',
    End_Year: 2022,
    Discipline: 'Microbiology',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    Resources: [ 'AI Workstations', '3D Printers', 'High-Performance Computers' ],
    Num_Personnel: 26,
    Project_Value: 2000000,
    Progress_Report: 'Quarterly ',
    University: {
      Name: 'University College Cork',
      City: 'Cork',
      Num_Undergrad: 17000,
      Specialism: 'Data Science',
      Annual_Budget: 78497
    },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Qualification: 'PhD',
      Salary: '81000',
      Specialism: 'Artificial Intelligence',
      Num_Publication: 22
    }
  }
]
researchDB> |
```


Mongo Shell: Simple Queries

Objective:

Demonstrate a wide range of retrieval (READ) queries using different MongoDB query selectors. This section showcases basic to intermediate querying capabilities using the MongoDB Shell (`mongosh`). All queries focus solely on the R in CRUD — *Retrieving* data from the collection, and exclude any creation, update, or deletion commands.

Approach:

Queries are divided by complexity level:

- Beginner (Queries 1–7): Simple queries using basic selectors and filters
- Intermediate (Queries 8–20): Combining multiple filters, embedded fields, regex, and arrays

Note: Queries 21–25 are not included here because they are advanced queries built using the Aggregation Framework and are covered in Stage 3 of this project.

Each query includes:

- A clear, well-written question in plain English
- The exact MongoDB shell command
- A matching `countDocuments()` version of the query for validation
- Targeted field projection to minimize output clutter
- Occasional use of `.pretty()` for output readability (e.g., in screencasts)

Best Practices Followed:

- Avoided using `.limit()` except once in the entire set
- Ensured every query includes at least three meaningful criteria
- Filtered for outputs returning fewer than 15 documents where feasible
- Demonstrated diverse selector usage (`$gt`, `$in`, `$regex`, etc.)

Beginner Query 1:

Question: Find all research projects with a value of exactly €500,000, that started in 2022, and belong to the "Biology" discipline.

```
db.projects.find(
  {
    Project_Value: 500000,
    Start_Year: "2022",
    Discipline: "Biology"
  },
  {
    Description: 1,
    Start_Year: 1,
    Discipline: 1,
    Project_Value: 1,
    _id: 0
  }
)
```

```
...
[
  {
    Description: 'Theoretical Approaches to Black Hole Thermodynamics',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'An Investigation into Plastic Degradation in Space Science',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'A Comprehensive Analysis of AI Algorithms',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'Development and Validation of Quantum Systems in Climate Change',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'An Investigation into Cybersecurity Threats',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'Innovative Use of Carbon Capture in Artificial Intelligence',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'An In-Depth Study on Plastic Degradation',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'Enhancing the Efficiency of Quantum Systems in Artificial Intelligence',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'Real-Time Monitoring and Evaluation of Genetic Variation in Artificial Intelligence',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  },
  {
    Description: 'Designing and Prototyping of Renewable Resources in Renewable Energy',
    Start_Year: 2022,
    Discipline: 'Biology',
    Project_Value: 500000
  }
]
researchDB> |
```

```
db.projects.countDocuments({
  Project_Value: 500000,
  Start_Year: 2022,
  Discipline: "Biology"
})
```

```
researchDB> db.projects.countDocuments({
...   Project_Value: 500000,
...   Start_Year: 2022,
...   Discipline: "Biology"
... })
...
10
researchDB> |
```

Explanation of Functions and Relevance

- **find():** This is the core MongoDB retrieval function. It searches the projects collection for documents that match **three specific criteria**:
 - Project_Value = 500000
 - Start_Year = "2022" (*Note: string, not number, based on your dataset*)
 - Discipline = "Biology"
- **Field Projection ({ Description: 1, ... }):** Limits the output to only the relevant fields — **Description, Start_Year, Discipline, and Project_Value**. This avoids clutter and makes the result easier to read and present.
- **countDocuments():** Immediately after running the main query, this function is used to return the number of matching documents. It's relevant for understanding how many results were returned by the filter — helps with "spot-checking" for accuracy.

Why It's Relevant to the Question

This query demonstrates **basic retrieval filtering using multiple fields**, and:

- Shows understanding of filtering documents with exact matches on numeric and string values.
- Applies best practices in **field projection** to return clean, focused output.
- Introduces a **count query** to validate how many records match the filter — this is important for managing result volume and verifying query precision.

Beginner Query 2 – Query on embedded fields

Question: Find projects led by scientists with a PhD, having over 25 publications, and working in the Climate Change field.

```
db.projects.find(
  {
    "Proj_Lead.Qualification": "PhD",
    "Proj_Lead.Num_Publication": { $gt: 25 },
    "Proj_Lead.Specialism": "Climate Change"
  },
  {
    Description: 1,
    "Proj_Lead.Name": 1,
    "Proj_Lead.Specialism": 1,
    "Proj_Lead.Num_Publication": 1,
    _id: 0
  }
)
```

```

{
  Description: 'Big Data Analytics in Monitoring Global Biodiversity',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Sustainable Methods for Synthesizing Biodegradable Polymers',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Leveraging Generative Adversarial Networks (GANs) for Data Augmentation in Medical Imaging',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Theoretical Approaches to Black Hole Thermodynamics',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Bioremediation of Oil Spills Using Microbial Communities',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Understanding the Thermodynamics of Solar Cells',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Blockchain Technology for Securing Biomedical Data',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
{
  Description: 'Design and Synthesis of Biodegradable Polymers for Environmental Applications',
  Proj_Lead: {
    Name: 'Dr. Aisha Rahman',
    Specialism: 'Climate Change',
    Num_Publication: 28
  }
},
},

```

```
db.projects.countDocuments({
  "Proj_Lead.Qualification": "PhD",
  "Proj_Lead.Num_Publication": { $gt: 25 },
  "Proj_Lead.Specialism": "Climate Change"
})
```

```
researchDB> db.projects.countDocuments({
...   "Proj_Lead.Qualification": "PhD",
...   "Proj_Lead.Num_Publication": { $gt: 25 },
...   "Proj_Lead.Specialism": "Climate Change"
... })
...
128
researchDB> |
```

Explanation of Functions and Relevance

- **Query on Embedded Fields:** This query filters fields *within the Proj_Lead embedded document*. Accessing nested fields using dot notation (e.g. "Proj_Lead.Qualification") is a core skill in MongoDB when working with document structures.
- **Operators Used:**
 - \$gt: Selects documents where the Num_Publication is **greater than 25** — a numeric comparison operator.
 - Exact matches for Qualification = "PhD" and Specialism = "Climate Change".
- **Field Projection:**
 - Only key information is returned: **project description, name, specialism, and number of publications** of the project lead. This keeps the output concise and relevant.
- **countDocuments():**
 - Used again to verify how many documents match the filter — an essential practice for result validation and tracking.

Why It's Relevant to the Question

This query tests your ability to:

- Work with nested structures (embedded documents).
- Apply logical filters to select specific scientist attributes.
- Use numeric comparison operators like \$gt in real-world queries.
- Design queries that are accurate, efficient, and return a small, relevant result set — aligning with best practices for MongoDB shell querying.

Beginner Query 3 – Using \$in with arrays

Question: Find all projects sponsored by either the EU or WHO, focused on "Medicine", and running in Edinburgh.

```
db.projects.find(
  {
    Sponsor: { $in: [/\s*EU\s*$/, /\s*WHO\s*$/] },
    Subject_Area: { $regex: "^\s*Medicine\s*$", $options: "i" },
    "University.City": { $regex: "^\s*Edinburgh\s*$", $options: "i" }
  },
  {
    Description: 1,
    Sponsor: 1,
    Subject_Area: 1,
    "University.City": 1,
    _id: 0
  }
)
```

```
...
[
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Subject_Area: 'Medicine',
    Sponsor: [ 'EU ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Enhancing the Efficiency of Drug Discovery in Climate Change',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'An Investigation into Carbon Capture in Climate Change',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Enhancing the Efficiency of Epidemiological Patterns in Space Science',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Enhancing the Efficiency of Electric Vehicles in Space Science',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Designing and Prototyping of Plastic Degradation in Renewable Energy',
    Subject_Area: 'Medicine',
    Sponsor: [ 'EU ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Predictive Modeling for Data Privacy in Renewable Energy',
    Subject_Area: 'Medicine',
    Sponsor: [ 'WHO ' ],
    University: { City: 'Edinburgh' }
  }
]
researchDB> |
```

```
db.projects.countDocuments({
  Sponsor: { $in: [/^\s*EU\s*$/, /^\s*WHO\s*$/] },
  Subject_Area: { $regex: "^\s*Medicine\s*$", $options: "i" },
  "University.City": { $regex: "^\s*Edinburgh\s*$", $options: "i" }
})
```

```
researchDB> db.projects.countDocuments({
...   Sponsor: { $in: [/^\s*EU\s*$/, /^\s*WHO\s*$/] },
...   Subject_Area: { $regex: "^\s*Medicine\s*$", $options: "i" },
...   "University.City": { $regex: "^\s*Edinburgh\s*$", $options: "i" }
... })
...
7
researchDB> |
```

Explanation of Functions and Relevance

- **\$in Operator:** Used here to search documents where the Sponsor array contains **either "EU" or "WHO"**. This is useful when the field is an array and we want to match any of several values.
- **\$regex:**
 - Used to match values even if they include inconsistent whitespace
 - Applied to both Subject_Area and University.City to ensure flexible and accurate string matching
 - The `^\s*...\s*$` pattern ensures we match full values even if padded with leading/trailing spaces
- **Field Projection:**
 - Only shows relevant fields: Description, Sponsor, Subject_Area, and City, keeping the output clean and readable
- **countDocuments():**
 - Essential for showing how many records meet the conditions — useful for validation and ensuring the query returns a manageable result set

Why It's Relevant to the Question

This query:

- Demonstrates how to match array values using \$in with regular expressions
- Showcases robust string filtering using \$regex for real-world messy data
- Reflects typical query logic in scientific research projects where sponsors may vary, and multiple filtering dimensions (e.g., location and topic) are needed

Beginner Query 4 – Logical AND/OR Combination

Question: Find projects either in "Physics" or "Biology", started in 2021, and funded exactly €1,000,000.

```
db.projects.find(
  {
    Num_Personnel: { $gt: 20 },
    Subject_Area: { $regex: "^\\s*Renewable Energy\\s*$", $options: "i" },
    "University.City": { $regex: "^\\s*Cambridge\\s*$", $options: "i" }
  },
  {
    Description: 1,
    Num_Personnel: 1,
    Subject_Area: 1,
    "University.City": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Photocatalytic Water Splitting for Sustainable Energy',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 22,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Forecasting Patterns in Nutrient Cycling in Space Science',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 31,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Designing and Prototyping of Electric Vehicles in Artificial Intelligence',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 37,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Predictive Modeling for Quantum Systems in Climate Change',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 31,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Enhancing the Efficiency of Epidemiological Patterns in Artificial Intelligence',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 35,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Innovative Use of Battery Efficiency in Space Science',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 25,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Designing and Prototyping of Solar Energy in Renewable Energy',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 25,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'An Investigation into Cybersecurity Threats in Artificial Intelligence',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 34,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'A Comprehensive Analysis of Microbial Growth in Space Science',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 22,
    University: { City: 'Cambridge' }
  },
  {
    Description: 'An In-Depth Study on Plastic Degradation in Renewable Energy',
    Subject_Area: 'Renewable Energy',
    Num_Personnel: 38,
    University: { City: 'Cambridge' }
  },
]
```

```
db.projects.countDocuments({
  Num_Personnel: { $gt: 20 },
  Subject_Area: { $regex: "^\\s*Renewable Energy\\s*$", $options: "i" },
  "University.City": { $regex: "^\\s*Cambridge\\s*$", $options: "i" }
})
```

```
researchDB> db.projects.countDocuments({
...   Num_Personnel: { $gt: 20 },
...   Subject_Area: { $regex: "^\\s*Renewable Energy\\s*$", $options: "i" },
...   "University.City": { $regex: "^\\s*Cambridge\\s*$", $options: "i" }
... })
11
researchDB> |
```

Explanation of Functions and Relevance

- **\$all Operator:** Ensures both "High-Performance Computers" and "Quantum Lab" are present in the Resources array. Unlike \$in, \$all demands all listed elements match
- **Embedded Field Matching:** "Proj_Lead.Qualification" and "Proj_Lead.Specialism" access fields inside the embedded Proj_Lead document. This is essential when filtering based on the attributes of the person leading the project
- **\$in Operator:** Used here to check if the sponsor is the "Government of Ireland", which supports array matching even if sponsors are inconsistently spaced or multiple
- **Field Projection:** Returns only relevant fields like Description, Resources, and Proj_Lead details, keeping output clean
- **countDocuments():** Used to validate the number of documents returned using the same filter — especially important for ensuring precise and manageable query results

Why It's Relevant to the Question

This query:

- Demonstrates how to check for multiple array values simultaneously using \$all
- Shows how to combine embedded document filtering with top-level fields
- Reflects real-world search logic in scientific data where projects span multiple resource needs, involve expert researchers, and are publicly funded
- Helps decision-makers identify high-tech, government-backed research led by domain experts

Beginner Query 5 – Use of \$gte, \$lte, and \$or

Question:

Find all projects that started between 2020 and 2022, have fewer than 10 personnel, or are led by a project leader with fewer than 20 publications.

```
db.projects.find(
{
  Start_Year: { $gte: 2020, $lte: 2022 },
  $or: [
    { Num_Personnel: { $lt: 10 } },
    { "Proj_Lead.Num_Publication": { $lt: 20 } }
  ]
},
{
  Description: 1,
  Start_Year: 1,
  Num_Personnel: 1,
  "Proj_Lead.Num_Publication": 1,
  _id: 0
})
```

```
[
  {
    Description: 'Electrochemical Methods for Hydrogen Production',
    Start_Year: 2020,
    Num_Personnel: 10,
    Proj_Lead: { Num_Publication: 18 }
  },
  {
    Description: 'Catalysis in Organic Synthesis for Pharmaceutical Development',
    Start_Year: 2021,
    Num_Personnel: 14,
    Proj_Lead: { Num_Publication: 17 }
  },
  {
    Description: 'Chemistry of Lithium-Ion Batteries for Electric Vehicles',
    Start_Year: 2021,
    Num_Personnel: 18,
    Proj_Lead: { Num_Publication: 18 }
  },
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Start_Year: 2022,
    Num_Personnel: 6,
    Proj_Lead: { Num_Publication: 22 }
  },
  {
    Description: 'The Role of Microbial Communities in Biodegradation of Plastic Waste',
    Start_Year: 2022,
    Num_Personnel: 5,
    Proj_Lead: { Num_Publication: 15 }
  },
  {
    Description: 'Cosmological Simulations of Dark Matter Distribution',
    Start_Year: 2021,
    Num_Personnel: 23,
    Proj_Lead: { Num_Publication: 17 }
  },
  {
    Description: 'Quantum Field Theories in High-Energy Physics',
    Start_Year: 2021,
    Num_Personnel: 12,
    Proj_Lead: { Num_Publication: 15 }
  },
  {
    Description: 'Exploring the Mechanisms of Photocatalysis for Solar-Driven Water Splitting',
    Start_Year: 2021,
    Num_Personnel: 7,
    Proj_Lead: { Num_Publication: 22 }
  },
  {
    Description: 'Development of Probiotic Formulations for Enhanced Gut Health and Disease Prevention',
    Start_Year: 2020,
    Num_Personnel: 7,
    Proj_Lead: { Num_Publication: 18 }
  },
  {
    Description: 'Machine Learning for Identifying Drug-Target Interactions',
    Start_Year: 2021,
```

```

db.projects.countDocuments(
{
  Start_Year: { $gte: 2020, $lte: 2022 },
  $or: [
    { Num_Personnel: { $lt: 10 } },
    { "Proj_Lead.Num_Publication": { $lt: 20 } }
  ]
}
)

```

```

researchDB> db.projects.countDocuments(
...   {
...     Start_Year: { $gte: 2020, $lte: 2022 },
...     $or: [
...       { Num_Personnel: { $lt: 10 } },
...       { "Proj_Lead.Num_Publication": { $lt: 20 } }
...     ]
...   }
... )
305
researchDB> |

```

Explanation of Functions and Relevance:

- **\$gte and \$lte:**Used to filter Start_Year between 2020 and 2022 inclusively
- **\$or:**Matches documents that satisfy at least one condition: fewer personnel or fewer publications
- **Embedded Document Access:**Accesses a nested field Proj_Lead.Num_Publication using dot notation
- **Field Projection:**Displays just enough fields to understand the context
- **countDocuments():** Confirms how many projects meet the logic conditions

Why It's Relevant: This mirrors real-world use cases where multiple flexible criteria determine selection, especially useful in managing team sizes or evaluating project leadership productivity

Beginner Query 6 – Combining \$in with Embedded and Top-Level Fields

Question:

Find projects from the “Physics” or “Chemistry” disciplines, led by PhD holders, and based in Cambridge.

```
db.projects.find(
  {
    Discipline: { $in: ["Physics", "Chemistry"] },
    "Proj_Lead.Qualification": "PhD",
    "University.City": "Cambridge"
  },
  {
    Description: 1,
    Discipline: 1,
    "Proj_Lead.Name": 1,
    "Proj_Lead.Qualification": 1,
    "University.City": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Big Data Analytics in Monitoring Global Biodiversity',
    Discipline: 'Chemistry',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. Aisha Rahman', Qualification: 'PhD' }
  },
  {
    Description: 'Leveraging Generative Adversarial Networks (GANs) for Data Augmentation in Medical Imaging',
    Discipline: 'Chemistry',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. Aisha Rahman', Qualification: 'PhD' }
  },
  {
    Description: 'Microbial Pathways for Bioethanol Production',
    Discipline: 'Physics',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. Aisha Rahman', Qualification: 'PhD' }
  },
  {
    Description: 'Computational Studies of Electromagnetic Wave Propagation',
    Discipline: 'Physics',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. Ravi Kapoor', Qualification: 'PhD' }
  },
  {
    Description: 'Investigating the Role of Spintronics in Data Storage Technologies',
    Discipline: 'Physics',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Prof. Sarah Lin', Qualification: 'PhD' }
  },
  {
    Description: 'Modeling Gravitational Lensing in Astrophysics',
    Discipline: 'Chemistry',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. James O'Connor', Qualification: 'PhD' }
  },
  {
    Description: 'The Role of AI in Climate Modeling Predicting and Mitigating Climate Change Impacts',
    Discipline: 'Chemistry',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. Ravi Kapoor', Qualification: 'PhD' }
  },
  {
    Description: 'Photocatalytic Water Splitting for Sustainable Energy',
    Discipline: 'Physics',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. James O'Connor', Qualification: 'PhD' }
  },
  {
    Description: 'An Investigation into Photon Behavior in Artificial Intelligence',
    Discipline: 'Chemistry',
    University: { City: 'Cambridge' },
    Proj_Lead: { Name: 'Dr. James O'Connor', Qualification: 'PhD' }
  }
]
```

```

db.projects.countDocuments(
{
  Discipline: { $in: ["Physics", "Chemistry"] },
  "Proj_Lead.Qualification": "PhD",
  "University.City": "Cambridge"
}
)

```

```

researchDB> db.projects.countDocuments(
...   {
...     Discipline: { $in: ["Physics", "Chemistry"] },
...     "Proj_Lead.Qualification": "PhD",
...     "University.City": "Cambridge"
...   }
... )
32
researchDB> |

```

Explanation of Functions and Relevance:

- **\$in Operator:** Selects projects where Discipline is either "Physics" or "Chemistry"
- **Dot Notation:** Used to query and project embedded fields like Proj_Lead.Qualification
- **Field Projection:** Ensures clarity by displaying specific identifiers only
- **countDocuments():** Validates the size of result set before review

Why It's Relevant: Useful for institutions wanting to identify high-calibre, subject-specific projects under elite leadership

Beginner Query 7 – Use of \$exists and \$regex for Quality Assurance

Question:

Find projects with explicitly defined progress reports that include the word “Quarterly” (case-insensitive), and ensure they include a Sponsor field.

```
db.projects.find(
{
  Progress_Report: { $regex: "quarterly", $options: "i" },
  Sponsor: { $exists: true }
},
{
  Description: 1,
  Progress_Report: 1,
  Sponsor: 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Leveraging Quantum Computing for Understanding Protein Folding Mechanisms in Biological Systems',
    Sponsor: [ 'WHO ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Electrochemical Methods for Hydrogen Production',
    Sponsor: [ 'Government of Ireland ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Big Data Analytics in Monitoring Global Biodiversity',
    Sponsor: [ 'EU ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Sustainable Methods for Synthesizing Biodegradable Polymers',
    Sponsor: [ 'Government of Ireland ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Adaptation Mechanisms in Marine Organisms Facing Acidification',
    Sponsor: [ 'WHO ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Leveraging Generative Adversarial Networks (GANs) for Data Augmentation in Medical Imaging',
    Sponsor: [ 'EU ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Role of Microbial Consortia in Bioremediation of Heavy Metals',
    Sponsor: [ 'Department of Education ' ],
    Progress_Report: 'Quarterly '
  },
  {
    Description: 'Cosmological Simulations of Dark Matter Distribution',
    Sponsor: [ 'EU ' ],
    Progress_Report: 'Quarterly '
  },
  {

```

```
db.projects.countDocuments(  
  {  
    Progress_Report: { $regex: "quarterly", $options: "i" },  
    Sponsor: { $exists: true }  
  }  
)
```

```
researchDB> db.projects.countDocuments(  
...   {  
...     Progress_Report: { $regex: "quarterly", $options: "i" },  
...     Sponsor: { $exists: true }  
...   }  
... )  
326  
researchDB> |
```

Explanation of Functions and Relevance:

- **\$regex:** Used for flexible text matching regardless of case or spacing
- **\$exists:** Confirms the existence of the Sponsor array, ensuring no missing data
- **Field Projection:** Keeps output to only what's needed for validation and review
- **countDocuments():** Assesses how many projects are actively reporting progress quarterly

Why It's Relevant: Crucial for progress tracking and funding validation—ensures project documentation and sponsorship aren't missing or malformed

Intermediate Query 8 – Combining \$gt, \$regex, and Embedded Fields

Question: Find projects led by scientists earning over €75,000, working in “Artificial Intelligence”, and based at a university with more than 15,000 undergraduates

```
db.projects.find(
  {
    "Proj_Lead.Salary": { $gt: 75000 },
    "Proj_Lead.Specialism": { $regex: /^\\s*Artificial Intelligence\\s*$ /i },
    "University.Num_Undergrad": { $gt: 15000 }
  },
  {
    Description: 1,
    "Proj_Lead.Name": 1,
    "Proj_Lead.Salary": 1,
    "Proj_Lead.Specialism": 1,
    "University.Num_Undergrad": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Leveraging Quantum Computing for Understanding Protein Folding Mechanisms in Biological Systems',
    University: { Num_Undergrad: 17000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    University: { Num_Undergrad: 23000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'AI-Driven Simulations for Drug Discovery in Chemistry',
    University: { Num_Undergrad: 24000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'Innovative Catalysts for Green Applications',
    University: { Num_Undergrad: 17000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'Advancements in Polymer for Biodegradable Plastics',
    University: { Num_Undergrad: 23000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'Exploring the Mechanisms of Photocatalysis for Solar-Driven Water Splitting',
    University: { Num_Undergrad: 23000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  },
  {
    Description: 'Investigating Magnetoresistance in Advanced Materials',
    University: { Num_Undergrad: 20000 },
    Proj_Lead: {
      Name: 'Dr. Ravi Kapoor',
      Salary: 81000,
      Specialism: 'Artificial Intelligence'
    }
  }
],
```

```

db.projects.countDocuments({
  Resources: { $in: [/^s*Quantum Lab\s*$/, /^s*Spectrometers\s*$/] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 400000 }
})

```

```

researchDB> db.projects.countDocuments({
...   Resources: { $in: [/^s*Quantum Lab\s*$/, /^s*Spectrometers\s*$/] },
...   Num_Personnel: { $lt: 10 },
...   Project_Value: { $gt: 400000 }
... })
98
researchDB> |

```

Explanation of Functions and Relevance

- **\$gt (Greater Than):** Used to filter for scientists with salaries above €75,000. This helps narrow down to higher-paid researchers
- **\$regex with /^s*...s*/i:** Ensures string matching ignores leading/trailing spaces and case (e.g., “artificial intelligence ” matches “Artificial Intelligence”). Enhances accuracy in real-world data
- **Embedded Field Access (Proj_Lead.Salary, University.Num_Undergrad):** Enables filtering and retrieving nested data inside documents, critical for deeply structured JSON documents
- **Field Projection:** Only relevant fields like project description, salary, and university size are shown, keeping the output concise
- **countDocuments():** Provides the number of documents that meet all criteria, helpful for result validation and demonstrating efficiency of filtering

Why It’s Relevant to the Question

- This query evaluates cross-field conditions: one on the embedded project leader object and another on the university document
- It combines numerical, text, and nested field filtering to simulate a real-world scenario of identifying elite, well-staffed AI-led university research
- It reflects a typical analysis task in data analytics—identifying top-performing projects based on funding, personnel, and institutional context

Intermediate Query 9 – Query with \$in and \$lt

Question: Find projects that use either “Quantum Lab” or “Spectrometers” as resources, employ fewer than 10 personnel, and have a project value above €400,000.

```
db.projects.find(
{
  Resources: { $in: [/^s*Quantum Lab\s*$/, /^s*Spectrometers\s*$/] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 400000 }
},
{
  Description: 1,
  Resources: 1,
  Num_Personnel: 1,
  Project_Value: 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Resources: [
      'Cloud Computing Credits',
      'High-Performance Computers',
      'Spectrometers'
    ],
    Num_Personnel: 6,
    Project_Value: 500000
  },
  {
    Description: 'The Role of Microbial Communities in Biodegradation of Plastic Waste',
    Resources: [ 'Microscopes', '3D Printers', 'Spectrometers' ],
    Num_Personnel: 5,
    Project_Value: 500000
  },
  {
    Description: 'Innovative Catalysts for Green Applications',
    Resources: [ 'AI Workstations', 'Spectrometers', 'Cloud Computing Credits' ],
    Num_Personnel: 6,
    Project_Value: 1000000
  },
  {
    Description: 'Exploring the Mechanisms of Photocatalysis for Solar-Driven Water Splitting',
    Resources: [ 'Spectrometers', '3D Printers', 'Quantum Lab' ],
    Num_Personnel: 7,
    Project_Value: 1000000
  },
  {
    Description: 'AI for Personalized Medicine From Data to Decisions',
    Resources: [ 'Quantum Lab', 'AI Workstations', 'Laptops' ],
    Num_Personnel: 5,
    Project_Value: 2000000
  },
  {
    Description: 'Cryptography in Quantum Computing Networks',
    Resources: [ 'Laptops', 'Cloud Computing Credits', 'Quantum Lab' ],
    Num_Personnel: 6,
    Project_Value: 1000000
  },
  {
    Description: 'Advances in Understanding Microbial Genome Evolution',
    Resources: [ 'High-Performance Computers', 'Quantum Lab', 'Laptops' ],
    Num_Personnel: 9,
    Project_Value: 500000
  },
  {
    Description: 'Microbial Pathways for Bioethanol Production',
    Resources: [ 'Spectrometers', 'Laptops', 'Microscopes' ],
    Num_Personnel: 6,
    Project_Value: 500000
  },
  {
    Description: 'The Role of Gut Microbiota in Human Health and Disease',
    Resources: [ 'AI Workstations', 'Spectrometers', 'Quantum Lab' ],
    Num_Personnel: 5,
    Project_Value: 500000
  },
]
```

```
db.projects.countDocuments({
  Resources: { $in: [/^\\s*Quantum Lab\\s*$/, /^\\s*Spectrometers\\s*$/] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 400000 }
})
```

```
researchDB> db.projects.countDocuments({
...   Resources: { $in: [/^\\s*Quantum Lab\\s*$/, /^\\s*Spectrometers\\s*$/] },
...   Num_Personnel: { $lt: 10 },
...   Project_Value: { $gt: 400000 }
... })
98
researchDB> |
```

Explanation of Functions and Relevance:

- **\$in with \$regex:** Matches documents where the Resources array contains any of the specified equipment, accounting for potential whitespace
- **\$lt / \$gt Operators:** Filters by numerical values to select small teams with high-value projects
- **Field Projection:** Shows only relevant project attributes to reduce clutter
- **countDocuments():** Helps validate the number of matched documents
-

Why It's Relevant:

Useful in analysing high-tech, small-team research setups using expensive or advanced instruments

Intermediate Query 10 – Filtering with \$in, \$lt, and \$gt

Question: Find projects using either "Quantum Lab" or "Spectrometers" as resources, with fewer than 10 personnel and a project value above €400,000.

```
db.projects.find(
{
  Resources: { $in: [/^s*Quantum Lab\s*$/ , /^s*Spectrometers\s*$/] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 400000 }
},
{
  Description: 1,
  Resources: 1,
  Num_Personnel: 1,
  Project_Value: 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Resources: [
      'Cloud Computing Credits',
      'High-Performance Computers',
      'Spectrometers'
    ],
    Num_Personnel: 6,
    Project_Value: 500000
  },
  {
    Description: 'The Role of Microbial Communities in Biodegradation of Plastic Waste',
    Resources: [ 'Microscopes', '3D Printers', 'Spectrometers' ],
    Num_Personnel: 5,
    Project_Value: 500000
  },
  {
    Description: 'Innovative Catalysts for Green Applications',
    Resources: [ 'AI Workstations', 'Spectrometers', 'Cloud Computing Credits' ],
    Num_Personnel: 6,
    Project_Value: 1000000
  },
  {
    Description: 'Exploring the Mechanisms of Photocatalysis for Solar-Driven Water Splitting',
    Resources: [ 'Spectrometers', '3D Printers', 'Quantum Lab' ],
    Num_Personnel: 7,
    Project_Value: 1000000
  },
  {
    Description: 'AI for Personalized Medicine From Data to Decisions',
    Resources: [ 'Quantum Lab', 'AI Workstations', 'Laptops' ],
    Num_Personnel: 5,
    Project_Value: 2000000
  },
  {
    Description: 'Cryptography in Quantum Computing Networks',
    Resources: [ 'Laptops', 'Cloud Computing Credits', 'Quantum Lab' ],
    Num_Personnel: 6,
    Project_Value: 1000000
  },
  {
    Description: 'Advances in Understanding Microbial Genome Evolution',
    Resources: [ 'High-Performance Computers', 'Quantum Lab', 'Laptops' ],
    Num_Personnel: 9,
    Project_Value: 500000
  },
  {
    Description: 'Microbial Pathways for Bioethanol Production',
    Resources: [ 'Spectrometers', 'Laptops', 'Microscopes' ],
    Num_Personnel: 6,
    Project_Value: 500000
  },
  {
    Description: 'The Role of Gut Microbiota in Human Health and Disease',
    Resources: [ 'AI Workstations', 'Spectrometers', 'Quantum Lab' ],
    Num_Personnel: 5,
    Project_Value: 500000
  }
]
```

```
db.projects.countDocuments({
  Resources: { $in: [/^s*Quantum Lab\s*$/, /^s*Spectrometers\s*$/] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 400000 }
})
```

```
researchDB> db.projects.countDocuments({
...   Resources: { $in: [/^s*Quantum Lab\s*$/, /^s*Spectrometers\s*$/] },
...   Num_Personnel: { $lt: 10 },
...   Project_Value: { $gt: 400000 }
... })
98
researchDB> |
```

Explanation of Functions and Relevance:

- **\$in**: Selects documents containing any of the listed resources
- **\$lt**: Ensures smaller team sizes (under 10 personnel)
- **\$gt**: Focuses on high-value projects.
- **\$regex** inside **\$in**: Handles messy whitespace in array values
- **Field Projection**: Targets only essential fields to reduce clutter
- **countDocuments()**: Verifies result set size for manageable output

Why It's Relevant: This query is useful to identify elite small-scale research setups that handle expensive tech resources, useful for cost-efficiency or lab specialization insights

Intermediate Query 11 – Using \$all with Arrays

Question: Find projects that use all the following resources: "Laptops", "High-Performance Computers", and "Quantum Lab"

```
db.projects.find(
{
  Resources: { $all: ["Laptops", "High-Performance Computers", "Quantum Lab"] }
},
{
  Description: 1,
  Resources: 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Theoretical Approaches to Black Hole Thermodynamics',
    Resources: [ 'High-Performance Computers', 'Laptops', 'Quantum Lab' ]
  },
  {
    Description: 'Leveraging Generative Adversarial Networks (GANs) for Data Augmentation in Medical Imaging',
    Resources: [ 'Quantum Lab', 'Laptops', 'High-Performance Computers' ]
  },
  {
    Description: 'Machine Learning for Identifying Drug-Target Interactions',
    Resources: [ 'Laptops', 'High-Performance Computers', 'Quantum Lab' ]
  },
  {
    Description: 'The Dynamics of Binary Star Systems A Simulation Approach',
    Resources: [ 'Laptops', 'High-Performance Computers', 'Quantum Lab' ]
  },
  {
    Description: 'Advances in Understanding Microbial Genome Evolution',
    Resources: [ 'High-Performance Computers', 'Quantum Lab', 'Laptops' ]
  },
  {
    Description: 'Real-Time Monitoring and Evaluation of Microbial Growth in Artificial Intelligence',
    Resources: [ 'Quantum Lab', 'Laptops', 'High-Performance Computers' ]
  },
  {
    Description: 'An Investigation into Photon Behavior',
    Resources: [ 'High-Performance Computers', 'Laptops', 'Quantum Lab' ]
  },
  {
    Description: 'Advancing Strategies for Nutrient Cycling in Artificial Intelligence',
    Resources: [ 'High-Performance Computers', 'Laptops', 'Quantum Lab' ]
  },
  {
    Description: 'Innovative Use of Quantum Systems in Renewable Energy',
    Resources: [ 'High-Performance Computers', 'Laptops', 'Quantum Lab' ]
  },
  {
    Description: 'Investigating the Underlying Mechanisms of Bioinformatics in Space Science',
    Resources: [ 'Laptops', 'Quantum Lab', 'High-Performance Computers' ]
  },
  {
    Description: 'Development and Validation of Plastic Degradation in Climate Change',
    Resources: [ 'High-Performance Computers', 'Laptops', 'Quantum Lab' ]
  },
  {
    Description: 'Investigating the Underlying Mechanisms of Genetic Variation',
    Resources: [ 'Quantum Lab', 'High-Performance Computers', 'Laptops' ]
  },
  {
    Description: 'Real-Time Monitoring and Evaluation of Neural Networks in Artificial Intelligence',
    Resources: [ 'Quantum Lab', 'Laptops', 'High-Performance Computers' ]
  }
]
```

```
db.projects.countDocuments({
  Resources: { $all: ["Laptops", "High-Performance Computers", "Quantum Lab"] }
})
```

```
researchDB> db.projects.countDocuments({
...   Resources: { $all: ["Laptops", "High-Performance Computers", "Quantum Lab"] }
... })
23
researchDB> |
```

Explanation of Functions and Relevance

- **\$all Operator:** Ensures all specified elements must be present in the array. Perfect for checking full resource requirements.
- **Field Projection:** Limits output to relevant fields only — helpful when reviewing technical setups.
- **countDocuments():** Validates result count, especially when searching for rare combinations.
-

Why It's Relevant: This query models real-world scenarios where research projects require a specific suite of tools — ensuring completeness of infrastructure.

Intermediate Query 12 – Matching Nested and Numeric Criteria

Question: Retrieve projects based in “Dublin” that have over 25 personnel and are led by a PhD-qualified scientist.

```
db.projects.find(
{
  "University.City": { $regex: /\s*Dublin\s*$/i },
  Num_Personnel: { $gt: 25 },
  "Proj_Lead.Qualification": "PhD"
},
{
  Description: 1,
  Num_Personnel: 1,
  "University.City": 1,
  "Proj_Lead.Qualification": 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Theoretical Approaches to Black Hole Thermodynamics',
    Num_Personnel: 32,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'An In-Depth Study on Solar Energy in Space Science',
    Num_Personnel: 36,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Forecasting Patterns in Bioinformatics',
    Num_Personnel: 39,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Mathematical Modeling of Ocean Acidity',
    Num_Personnel: 33,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Forecasting Patterns in Solar Energy',
    Num_Personnel: 34,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'An In-Depth Study on Climate Models in Artificial Intelligence',
    Num_Personnel: 40,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Designing and Prototyping of Ocean Acidity in Climate Change',
    Num_Personnel: 31,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Simulation-Based Study on Cybersecurity Threats in Climate Change',
    Num_Personnel: 29,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Mathematical Modeling of Neural Networks in Renewable Energy',
    Num_Personnel: 40,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Predictive Modeling for Epidemiological Patterns in Artificial Intelligence',
    Num_Personnel: 27,
    University: { City: 'Dublin' },
    Proj_Lead: { Qualification: 'PhD' }
  },
]
```

```
db.projects.countDocuments({
  "University.City": { $regex: /\s*Dublin\s*/i },
  Num_Personnel: { $gt: 25 },
  "Proj_Lead.Qualification": "PhD"
})
```

```
researchDB> db.projects.countDocuments({
...   "University.City": { $regex: /\s*Dublin\s*/i },
...   Num_Personnel: { $gt: 25 },
...   "Proj_Lead.Qualification": "PhD"
... })
27
researchDB> |
```

Explanation of Functions and Relevance

- **\$regex:** Flexible city name matching, correcting inconsistent spacing/capitalization
- **\$gt:** Filters projects with significant team size
- **Field Projection:** Keeps the focus on staff size, leader qualifications, and location
- **countDocuments():** Validates if any such projects exist

Why It's Relevant: Ideal for analysing large-scale projects led by highly qualified personnel in a specific academic hub

Intermediate Query 13 – \$nin and \$not in Combination

Question: List projects not sponsored by "WHO" or "Private Donation", and not located in "Oxford".

```
db.projects.find(
{
  Sponsor: { $nin: [/\s*WHO\s*$/, /\s*Private Donation\s*$/] },
  "University.City": { $not: /\s*Oxford\s*$/i }
},
{
  Description: 1,
  Sponsor: 1,
  "University.City": 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Sponsor: [ 'EU ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Role of Microbial Consortia in Bioremediation of Heavy Metals',
    Sponsor: [ 'Department of Education ' ],
    University: { City: 'Glasgow' }
  },
  {
    Description: 'Electrochemical Methods for Hydrogen Production',
    Sponsor: [ 'Government of Ireland ' ],
    University: { City: 'Cardiff' }
  },
  {
    Description: 'Sustainable Methods for Synthesizing Biodegradable Polymers',
    Sponsor: [ 'Government of Ireland ' ],
    University: { City: 'Cardiff' }
  },
  {
    Description: 'Big Data Analytics in Monitoring Global Biodiversity',
    Sponsor: [ 'EU ' ],
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Leveraging Generative Adversarial Networks (GANs) for Data Augmentation in Medical Imaging',
    Sponsor: [ 'EU ' ],
    University: { City: 'Cambridge' }
  },
  {
    Description: 'Catalysis in Organic Synthesis for Pharmaceutical Development',
    Sponsor: [ 'Department of Education ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'Chemistry of Lithium-Ion Batteries for Electric Vehicles',
    Sponsor: [ 'Department of Education ' ],
    University: { City: 'Edinburgh' }
  },
  {
    Description: 'AI-Driven Simulations for Drug Discovery in Chemistry',
    Sponsor: [ 'IRC ' ],
    University: { City: 'Cardiff' }
  },
  {
    Description: 'Advancements in Federated Learning for Privacy-Preserving Machine Learning Models',
    Sponsor: [ 'Government of Ireland ' ],
    University: { City: 'Cardiff' }
  },
  {
    Description: 'Quantum Field Theories in High-Energy Physics',
    Sponsor: [ 'EU ' ],
    University: { City: 'Athlone' }
  }
]
```

```
db.projects.countDocuments({
```

```
Sponsor: { $nin: [/^s*WHO\s*$/, /^s*Private Donation\s*$/] },  
"University.City": { $not: /^s*Oxford\s*$/i }  
})
```

```
researchDB> db.projects.countDocuments({  
...   Sponsor: { $nin: [/^s*WHO\s*$/, /^s*Private Donation\s*$/] },  
...   "University.City": { $not: /^s*Oxford\s*$/i }  
... })  
610  
researchDB> |
```

Explanation of Functions and Relevance

- **\$nin:** Filters out unwanted sponsors from the array.
- **\$not + \$regex:** Excludes a city based on flexible text matching.
- **Field Projection:** Minimizes clutter, focuses on sponsor and location data.
- **countDocuments():** Confirms manageable output set.
-

Why It's Relevant: Useful for filtering out irrelevant sponsors and regions when evaluating unbiased or underrepresented projects.

Intermediate Query 14 – Combined Field Match and Resource Condition

Question: Show projects in "Scotland" using "Cloud Computing Credits" and having annual university budgets above €75,000.

```
db.projects.find(
{
  "Resources": { $in: [/^s*Cloud Computing Credits\s*$/] },
  "University.Annual_Budget": { $gt: 75000 },
  "University.City": { $in: [/Edinburgh/i, /Glasgow/i] }
},
{
  Description: 1,
  "University.City": 1,
  Resources: 1,
  "University.Annual_Budget": 1,
  _id: 0
})
```

```
[
  {
    Description: 'Microbial Resistance Mechanisms in Pathogenic Bacteria',
    Resources: [
      'Cloud Computing Credits',
      'High-Performance Computers',
      'Spectrometers'
    ],
    University: { City: 'Edinburgh', Annual_Budget: 84657 }
  },
  {
    Description: 'Design of Functional Nanomaterials for Energy Storage',
    Resources: [ 'Cloud Computing Credits', 'AI Workstations', 'Laptops' ],
    University: { City: 'Glasgow', Annual_Budget: 91571 }
  },
  {
    Description: 'Cryptography in Quantum Computing Networks',
    Resources: [ 'Laptops', 'Cloud Computing Credits', 'Quantum Lab' ],
    University: { City: 'Edinburgh', Annual_Budget: 90645 }
  },
  {
    Description: 'Exploring Nonlinear Dynamics in Fluid Mechanics',
    Resources: [ '3D Printers', 'Cloud Computing Credits', 'Microscopes' ],
    University: { City: 'Glasgow', Annual_Budget: 111350 }
  },
  {
    Description: 'Plant-Microbe Interactions in Agricultural Sustainability',
    Resources: [
      'Cloud Computing Credits',
      'High-Performance Computers',
      'Spectrometers'
    ],
    University: { City: 'Glasgow', Annual_Budget: 93530 }
  },
  {
    Description: 'AI Techniques for Molecular Docking Simulations',
    Resources: [ 'Laptops', 'Quantum Lab', 'Cloud Computing Credits' ],
    University: { City: 'Glasgow', Annual_Budget: 98616 }
  },
  {
    Description: 'AI-Powered Predictive Maintenance Systems in Industrial IoT Enhancing Efficiency and Reducing Downtime',
    Resources: [ 'Cloud Computing Credits', '3D Printers', 'AI Workstations' ],
    University: { City: 'Glasgow', Annual_Budget: 96077 }
  },
  {
    Description: 'Predictive Modeling for Plastic Degradation in Artificial Intelligence',
    Resources: [ 'Cloud Computing Credits', 'Quantum Lab', 'AI Workstations' ],
    University: { City: 'Glasgow', Annual_Budget: 92534 }
  },
  {
    Description: 'Enhancing the Efficiency of Photon Behavior in Renewable Energy',
    Resources: [
      'High-Performance Computers',
      'Laptops',
      'Cloud Computing Credits'
    ],
    University: { City: 'Glasgow', Annual_Budget: 104136 }
  },
  {
    Description: 'Investigating the Underlying Mechanisms of AI Algorithms',

```

```
db.projects.countDocuments({
  "Resources": { $in: [/^\s*Cloud Computing Credits\s*$/] },
  "University.Annual_Budget": { $gt: 75000 },
  "University.City": { $in: [/Edinburgh/i, /Glasgow/i] }
})
```

```
researchDB> db.projects.countDocuments({
...   "Resources": { $in: [/^\s*Cloud Computing Credits\s*$/] },
...   "University.Annual_Budget": { $gt: 75000 },
...   "University.City": { $in: [/Edinburgh/i, /Glasgow/i] }
... })
46
researchDB> |
```

Explanation of Functions and Relevance

- **\$in + \$regex:** Allows flexible value matching from a list (city and resource)
- **\$gt:** Captures universities with strong financial backing
- **Field Projection:** Displays relevant fiscal and location data only
- **countDocuments():** Essential for verifying expected project volume
-

Why It's Relevant: Targets advanced, well-funded Scottish institutions using cutting-edge computational tools

Intermediate Query 15 – Using \$or for Flexible Filtering

Question: Return projects in either "Biology" or "Chemistry", with fewer than 10 personnel and ending before 2023.

```
db.projects.find(
  {
    $or: [
      { Discipline: { $regex: /\s*Biology\s*$/i } },
      { Discipline: { $regex: /\s*Chemistry\s*$/i } }
    ],
    Num_Personnel: { $lt: 10 },
    End_Year: { $lt: 2023 }
  },
  {
    Description: 1,
    Discipline: 1,
    Num_Personnel: 1,
    End_Year: 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Role of Microbial Consortia in Bioremediation of Heavy Metals',
    End_Year: 2022,
    Discipline: 'Chemistry',
    Num_Personnel: 7
  },
  {
    Description: 'Modeling Gravitational Lensing in Astrophysics',
    End_Year: 2021,
    Discipline: 'Chemistry',
    Num_Personnel: 5
  },
  {
    Description: 'Advancing Strategies for Quantum Systems in Climate Change',
    End_Year: 2021,
    Discipline: 'Biology',
    Num_Personnel: 6
  },
  {
    Description: 'Enhancing the Efficiency of Photon Behavior',
    End_Year: 2021,
    Discipline: 'Biology',
    Num_Personnel: 9
  },
  {
    Description: 'Enhancing the Efficiency of Microbial Growth in Renewable Energy',
    End_Year: 2022,
    Discipline: 'Chemistry',
    Num_Personnel: 7
  },
  {
    Description: 'Mathematical Modeling of Photon Behavior in Renewable Energy',
    End_Year: 2020,
    Discipline: 'Chemistry',
    Num_Personnel: 8
  },
  {
    Description: 'An In-Depth Study on Carbon Capture',
    End_Year: 2021,
    Discipline: 'Chemistry',
    Num_Personnel: 8
  },
  {
    Description: 'Forecasting Patterns in Epidemiological Patterns',
    End_Year: 2022,
    Discipline: 'Chemistry',
    Num_Personnel: 9
  },
  {
    Description: 'An In-Depth Study on Electric Vehicles',
    End_Year: 2022,
    Discipline: 'Biology',
    Num_Personnel: 5
  },
  {
    Description: 'Designing and Prototyping of Cybersecurity Threats in Renewable Energy',
    End_Year: 2022,
    Discipline: 'Biology',
    Num_Personnel: 6
  }
]
```

```

db.projects.countDocuments({
  $or: [
    { Discipline: { $regex: /\s*Biology\s*/i } },
    { Discipline: { $regex: /\s*Chemistry\s*/i } }
  ],
  Num_Personnel: { $lt: 10 },
  End_Year: { $lt: 2023 }
})

```

```

researchDB> db.projects.countDocuments({
...   $or: [
...     { Discipline: { $regex: /\s*Biology\s*/i } },
...     { Discipline: { $regex: /\s*Chemistry\s*/i } }
...   ],
...   Num_Personnel: { $lt: 10 },
...   End_Year: { $lt: 2023 }
... })
14
researchDB> |

```

Explanation of Functions and Relevance

- **\$or:** Enables multi-discipline filtering within a single condition
- **\$lt:** Used to narrow down personnel and timeline constraints
- **\$regex:** Handles inconsistent formatting in disciplines
- **countDocuments():** Important for managing output size
-

Why It's Relevant: Supports quick filtering of short-term, small-team STEM research across two key disciplines

Intermediate Query 16 – Using \$and for Explicit AND Logic

Question: Find projects funded by “Department of Education”, working in “Climate Change”, and conducted by scientists with over 20 publications.

```
db.projects.find(
  {
    $and: [
      { Sponsor: { $in: [/^\s*Department of Education\s*$/] } },
      { Subject_Area: { $regex: /^\s*Climate Change\s*$/i } },
      { "Proj_Lead.Num_Publication": { $gt: 20 } }
    ]
  },
  {
    Description: 1,
    Sponsor: 1,
    Subject_Area: 1,
    "Proj_Lead.Num_Publication": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'AI-Based Models for Predicting Climate Change Impacts',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 28 }
  },
  {
    Description: 'Metagenomics in Soil Microbial Diversity Studies',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 26 }
  },
  {
    Description: 'Simulation-Based Study on Genetic Variation in Artificial Intelligence',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 22 }
  },
  {
    Description: 'Simulation-Based Study on Microbial Growth in Artificial Intelligence',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 26 }
  },
  {
    Description: 'Innovative Use of Quantum Systems in Renewable Energy',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 22 }
  },
  {
    Description: 'Innovative Use of Nutrient Cycling',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 26 }
  },
  {
    Description: 'An In-Depth Study on Epidemiological Patterns',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 21 }
  },
  {
    Description: 'A Comprehensive Analysis of Nutrient Cycling in Artificial Intelligence',
    Subject_Area: 'Climate Change',
    Sponsor: [ 'Department of Education ' ],
    Proj_Lead: { Num_Publication: 26 }
  },
  {
  }
```

```

db.projects.countDocuments({
  $and: [
    { Sponsor: { $in: [/^\s*Department of Education\s*$/] } },
    { Subject_Area: { $regex: /^\s*Climate Change\s*$/i } },
    { "Proj_Lead.Num_Publication": { $gt: 20 } }
  ]
})

```

```

researchDB> db.projects.countDocuments({
...   $and: [
...     { Sponsor: { $in: [/^\s*Department of Education\s*$/] } },
...     { Subject_Area: { $regex: /^\s*Climate Change\s*$/i } },
...     { "Proj_Lead.Num_Publication": { $gt: 20 } }
...   ]
... })
16
researchDB> |

```

Explanation of Functions and Relevance

- **\$and:** Ensures all specified conditions are explicitly required.
- **\$in + \$regex:** Enables error-tolerant string matching in arrays.
- **\$gt:** Targets highly experienced project leaders.
- **countDocuments():** Confirms result set size before displaying full query output.

Why It's Relevant: Mimics grant evaluation logic where experience, subject matter, and funder alignment are critical.

Intermediate Query 17 – Using \$gte, \$lt, and array length filtering

Question:

Find projects with a value between €500,000 and €1,000,000, that use exactly 3 resources, and are in the field of "Climate Change".

```
db.projects.find(
{
  Project_Value: { $gte: 500000, $lt: 1000000 },
  Resources: { $size: 3 },
  Subject_Area: { $regex: /^s*Climate Change\s*$i }
},
{
  Description: 1,
  Project_Value: 1,
  Resources: 1,
  Subject_Area: 1,
  _id: 0
})
```

```
[
{
  Description: 'Adaptation Mechanisms in Marine Organisms Facing Acidification',
  Subject_Area: 'Climate Change',
  Resources: [
    'Microscopes',
    'Cloud Computing Credits',
    'High-Performance Computers'
  ],
  Project_Value: 500000
},
{
  Description: 'Big Data Analytics in Monitoring Global Biodiversity',
  Subject_Area: 'Climate Change',
  Resources: [ 'High-Performance Computers', '3D Printers', 'AI Workstations' ],
  Project_Value: 500000
},
{
  Description: 'Biotechnological Innovations in Microbial Fuel Cells',
  Subject_Area: 'Climate Change',
  Resources: [
    'Cloud Computing Credits',
    'High-Performance Computers',
    'Quantum Lab'
  ],
  Project_Value: 500000
},
{
  Description: 'Impact of Urbanization on Wildlife Population Dynamics',
  Subject_Area: 'Climate Change',
  Resources: [
    'High-Performance Computers',
    'AI Workstations',
    'Cloud Computing Credits'
  ],
  Project_Value: 500000
},
{
  Description: 'Exploring Extremophiles in Deep Sea Environments',
  Subject_Area: 'Climate Change',
  Resources: [ 'Laptops', 'AI Workstations', '3D Printers' ],
  Project_Value: 500000
},
{
  Description: 'Antibiotic Resistance in Aquatic Microbial Ecosystems',
  Subject_Area: 'Climate Change',
  Resources: [ 'Laptops', 'Spectrometers', 'Quantum Lab' ],
  Project_Value: 500000
},
{
  Description: 'Genetic Adaptations in Plants Under Climate Stress',
  Subject_Area: 'Climate Change',
  Resources: [ 'Microscopes', 'Cloud Computing Credits', 'Quantum Lab' ],
  Project_Value: 500000
},
]
```

```
db.projects.countDocuments({
  Project_Value: { $gte: 500000, $lt: 1000000 },
  Resources: { $size: 3 },
  Subject_Area: { $regex: /^s*Climate Change\s*$/i }
})
```

```
researchDB> db.projects.countDocuments({
...   Project_Value: { $gte: 500000, $lt: 1000000 },
...   Resources: { $size: 3 },
...   Subject_Area: { $regex: /^s*Climate Change\s*$/i }
... })
56
researchDB> |
```

Explanation of Functions and Relevance:

- **\$or Operator:** Allows flexible logic, matching documents that satisfy either of the two main conditions (low personnel or high value).
- **\$lt and \$gt:** Used for numeric comparisons on personnel and project value.
- **\$regex with i option:** Ensures we match “Medicine” despite inconsistent casing or whitespace.
- **Field Projection:** Returns only the fields required to verify results and keep output tidy.
-

Why It’s Relevant: This query helps identify unusually small teams or high-value projects, which might require extra oversight or represent flagship initiatives.

Intermediate Query 18 – Match on nested fields and numeric threshold

Question:

Identify projects at universities with budgets over €80,000, with more than 20 team members, and whose project lead has over 20 publications.

```
db.projects.find(
  {
    "University.Annual_Budget": { $gt: 80000 },
    Num_Personnel: { $gt: 20 },
    "Proj_Lead.Num_Publication": { $gt: 20 }
  },
  {
    Description: 1,
    "University.Annual_Budget": 1,
    Num_Personnel: 1,
    "Proj_Lead.Num_Publication": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Adaptation Mechanisms in Marine Organisms Facing Acidification',
    Num_Personnel: 26,
    University: { Annual_Budget: 88599 },
    Proj_Lead: { Num_Publication: 21 }
  },
  {
    Description: 'Advances in Theoretical Models of Black Hole Dynamics',
    Num_Personnel: 33,
    University: { Annual_Budget: 113082 },
    Proj_Lead: { Num_Publication: 26 }
  },
  {
    Description: 'Big Data Analytics in Monitoring Global Biodiversity',
    Num_Personnel: 37,
    University: { Annual_Budget: 92414 },
    Proj_Lead: { Num_Publication: 28 }
  },
  {
    Description: 'AI-Driven Simulations for Drug Discovery in Chemistry',
    Num_Personnel: 28,
    University: { Annual_Budget: 105657 },
    Proj_Lead: { Num_Publication: 22 }
  },
  {
    Description: 'CRISPR Applications in Combating Antimicrobial Resistance',
    Num_Personnel: 34,
    University: { Annual_Budget: 107704 },
    Proj_Lead: { Num_Publication: 21 }
  },
  {
    Description: 'Blockchain Technology for Securing Biomedical Data',
    Num_Personnel: 40,
    University: { Annual_Budget: 103832 },
    Proj_Lead: { Num_Publication: 28 }
  },
  {
    Description: 'Design of Functional Nanomaterials for Energy Storage',
    Num_Personnel: 25,
    University: { Annual_Budget: 91571 },
    Proj_Lead: { Num_Publication: 26 }
  },
  {
    Description: 'Quantum Computing in Cryptography',
    Num_Personnel: 30,
    University: { Annual_Budget: 120000 },
    Proj_Lead: { Num_Publication: 30 }
  }
]
```

```
db.projects.countDocuments({
  "University.Annual_Budget": { $gt: 80000 },
  Num_Personnel: { $gt: 20 },
  "Proj_Lead.Num_Publication": { $gt: 20 }
})
```

```
researchDB> db.projects.countDocuments({
...   "University.Annual_Budget": { $gt: 80000 },
...   Num_Personnel: { $gt: 20 },
...   "Proj_Lead.Num_Publication": { $gt: 20 }
... })
133
researchDB> |
```

Explanation of Functions and Relevance:

- **\$gt**: Enables threshold-based filtering on budgets, staff count, and publication volume
- **Field Projection**: Keeps results relevant and clear
- **Embedded Field Access**: Filters deeply nested fields like budget and publications

Why It's Relevant: It targets established, well-funded projects with experienced leads — a useful pattern for identifying impactful work or award-worthy programs

Intermediate Query 19 – Using \$in with nested and numeric constraints

Question:

Find projects located in Oxford or Dublin, with under 10 personnel, and project value above €600,000.

```
db.projects.find(
{
  "University.City": { $in: [/^\s*Oxford\s*$/i, /^\s*Dublin\s*$/i] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 600000 }
},
{
  Description: 1,
  "University.City": 1,
  Num_Personnel: 1,
  Project_Value: 1,
  _id: 0
}
)
```

```
[
  {
    Description: 'AI for Personalized Medicine From Data to Decisions',
    Num_Personnel: 5,
    Project_Value: 2000000,
    University: { City: 'Dublin' }
  },
  {
    Description: 'Enhancing the Efficiency of Drug Discovery',
    Num_Personnel: 7,
    Project_Value: 1000000,
    University: { City: 'Dublin' }
  },
  {
    Description: 'Real-Time Monitoring and Evaluation of Solar Energy',
    Num_Personnel: 7,
    Project_Value: 2000000,
    University: { City: 'Dublin' }
  },
  {
    Description: 'Investigating the Underlying Mechanisms of Ocean Acidity',
    Num_Personnel: 7,
    Project_Value: 2000000,
    University: { City: 'Oxford' }
  },
  {
    Description: 'Enhancing the Efficiency of Photon Behavior',
    Num_Personnel: 9,
    Project_Value: 1000000,
    University: { City: 'Oxford' }
  },
  {
    Description: 'Forecasting Patterns in Carbon Capture in Space Science',
    Num_Personnel: 9,
    Project_Value: 3000000,
    University: { City: 'Oxford' }
  },
  {
    Description: 'Mathematical Modeling of Drug Discovery in Space Science',
    Num_Personnel: 7,
    Project_Value: 2000000,
    University: { City: 'Dublin' }
  },
  {
    Description: 'Investigating the Underlying Mechanisms of AI Algorithms in Artificial Intelligence',
    Num_Personnel: 7,
    Project_Value: 2000000,
    University: { City: 'Dublin' }
  }
]
```

```
db.projects.countDocuments({
  "University.City": { $in: [/^\s*Oxford\s*$/i, /^\s*Dublin\s*$/i] },
  Num_Personnel: { $lt: 10 },
  Project_Value: { $gt: 600000 }
})
```

```
researchDB> db.projects.countDocuments({
...   "University.City": { $in: [/^\s*Oxford\s*$/i, /^\s*Dublin\s*$/i] },
...   Num_Personnel: { $lt: 10 },
...   Project_Value: { $gt: 600000 }
... })
30
researchDB> |
```

Explanation of Functions and Relevance:

- **\$in with \$regex:** Matches either Oxford or Dublin with whitespace/case flexibility
- **\$lt and \$gt:** Numeric range filters on team size and funding
- **Projection:** Clean display for geographical and resource-related analysis
-

Why It's Relevant: Targets high-value, low-personnel projects in leading academic hubs — ideal for cost-efficiency analysis or review of innovative pilot projects

Intermediate Query 20 – Combined \$regex on multiple text fields

Question:

List projects focused on "Artificial Intelligence" at universities with a specialism in "Data Science" and led by someone with a "PhD".

```
db.projects.find(
  {
    Subject_Area: { $regex: /\s*Artificial Intelligence\s*$/i },
    "University.Specialism": { $regex: /\s*Data Science\s*$/i },
    "Proj_Lead.Qualification": "PhD"
  },
  {
    Description: 1,
    Subject_Area: 1,
    "University.Specialism": 1,
    "Proj_Lead.Qualification": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'Investigating the Underlying Mechanisms of Photon Behavior in Climate Change',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Real-Time Monitoring and Evaluation of Climate Models in Renewable Energy',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'An Investigation into Drug Discovery in Space Science',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Innovative Use of Climate Models in Renewable Energy',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Forecasting Patterns in Ocean Acidity in Space Science',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Assessment of Carbon Capture in Space Science',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {
    Description: 'Development and Validation of Ocean Acidity in Renewable Energy',
    Subject_Area: 'Artificial Intelligence',
    University: { Specialism: 'Data Science' },
    Proj_Lead: { Qualification: 'PhD' }
  },
  {

```

```
db.projects.countDocuments({
  Subject_Area: { $regex: /\s*Artificial Intelligence\s*/i },
  "University.Specialism": { $regex: /\s*Data Science\s*/i },
  "Proj_Lead.Qualification": "PhD"
})
```

```
researchDB> db.projects.countDocuments({
...   Subject_Area: { $regex: /\s*Artificial Intelligence\s*/i },
...   "University.Specialism": { $regex: /\s*Data Science\s*/i },
...   "Proj_Lead.Qualification": "PhD"
... })
16
researchDB> |
```

Explanation of Functions and Relevance:

- **\$regex:** Ensures flexibility in string-matching — helpful in inconsistent datasets
- **Embedded Field Filtering:** Matches on deep fields like university and lead details
- **Projection:** Tailored to question focus without clutter

Why It's Relevant: Supports focused retrieval of cutting-edge tech projects — essential for exploring the AI research landscape or highlighting cross-disciplinary work

Intermediate Query 21

Find projects that started between 2020 and 2023 (inclusive), led by scientists specializing in *Space Science*, and where the university has a budget of less than €90,000.

```
db.projects.find(
  {
    Start_Year: { $gte: 2020, $lte: 2023 },
    "Proj_Lead.Specialism": { $regex: /\s*Space Science\s*/i },
    "University.Annual_Budget": { $lt: 90000 }
  },
  {
    Description: 1,
    Start_Year: 1,
    "Proj_Lead.Name": 1,
    "Proj_Lead.Specialism": 1,
    "University.Name": 1,
    "University.Annual_Budget": 1,
    _id: 0
  }
)
```

```
[
  {
    Description: 'The Role of Microbial Communities in Biodegradation of Plastic Waste',
    Start_Year: 2022,
    University: { Name: 'University College Cork', Annual_Budget: 46663 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Quantum Field Theories in High-Energy Physics',
    Start_Year: 2021,
    University: { Name: 'TUS: Midlands Midwest', Annual_Budget: 72183 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Exploring the Synthesis and Application of Metal-Organic Frameworks (MOFs) for Carbon Capture',
    Start_Year: 2020,
    University: { Name: 'University College Cork', Annual_Budget: 61066 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Natural Language Processing for Analyzing Scientific Literature',
    Start_Year: 2020,
    University: { Name: 'TUS: Midlands Midwest', Annual_Budget: 54667 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Genomics of Invasive Species and Their Environmental Impact',
    Start_Year: 2022,
    University: { Name: 'TUS: Midlands Midwest', Annual_Budget: 49152 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Predictive Modeling for Nutrient Cycling in Space Science',
    Start_Year: 2020,
    University: { Name: 'University of Oxford', Annual_Budget: 55923 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {
    Description: 'Advancing Strategies for AI Algorithms',
    Start_Year: 2022,
    University: { Name: 'University of Oxford', Annual_Budget: 46847 },
    Proj_Lead: { Name: 'Prof. Liam Byrne', Specialism: 'Space Science' }
  },
  {

```

```

db.projects.countDocuments(
{
  Start_Year: { $gte: 2020, $lte: 2023 },
  "Proj_Lead.Specialism": { $regex: "\\bSpace Science\\b", $options: "i" },
  "University.Annual_Budget": { $lt: 90000 }
}
)
researchDB> db.projects.countDocuments(
... {
...   Start_Year: { $gte: 2020, $lte: 2023 },
...   "Proj_Lead.Specialism": { $regex: "\\bSpace Science\\b", $options: "i" },
...   "University.Annual_Budget": { $lt: 90000 }
... }
... )
...
55
researchDB> |

```

Explanation of Functions and Relevance

- **\$gte / \$lte Operators:** Used to filter projects that started between 2020 and 2023, inclusive. These operators perform numeric comparisons to define a specific date range. Ideal for trend analysis, funding cycles, or temporal project evaluations
- **\$regex with i option:** Enables case-insensitive matching of "Environmental Science" in the Proj_Lead.Specialism field. Helps account for inconsistent casing or extra spaces in real-world datasets
- **\$lt Operator:** Filters projects from universities with lower budgets (less than €90,000). This is useful for assessing resource-constrained institutions or studying budget-performance correlations
- **Embedded Field Filtering:** Targets fields nested within subdocuments (like Proj_Lead.Specialism and University.Annual_Budget) — demonstrating how to query deeply structured JSON data
- **Field Projection:** Limits output to essential fields: Description, Start_Year, Proj_Lead.Specialism, University.Name, and Annual_Budget. This ensures results are clear and uncluttered
- **countDocuments():** Runs directly after the main query using the same filter to return the total number of matching documents. Critical for confirming that the query is efficient and returning expected results

Why It's Relevant to the Question

This query demonstrates how to:

- Extract environment-related projects during a specific time window.
- Cross-check if they were led by scientists in environmental fields.
- See which projects were conducted in universities with smaller budgets — which could point to low-resource innovation or regional sustainability efforts.

This is highly applicable in research funding, environmental impact studies, and policy planning.

Mongo Shell: Aggregation Framework

Objective:

Demonstrate use of the MongoDB Aggregation Framework — a powerful tool for performing complex data transformations, filtering, and summarization. This section mirrors the role of `GROUP BY`, `HAVING`, and `JOIN` in SQL, and is the core of analytical work in MongoDB.

Approach:

Includes 12 expert-level queries (Queries 21–32) using the `$aggregate` pipeline. These queries build upon the previous 20 simple queries and focus on complex summarisation and analytical insights.

Each query includes:

- A clearly defined analytical question
- A well-formatted, multi-stage aggregation pipeline (each stage on a new line)
- Multiple filtering criteria and thoughtful projection
- Justification for why the query adds value to the dataset
- Explanation focused on newly introduced functions/operators in each case

Best Practices Followed:

- Output limited to fewer than 15 documents where possible
- Every query adds analytical or strategic value
- Covers multiple aggregation operators such as `$group`, `$project`, `$unwind`, `$match`, `$sort`, `$count`, `$bucket`, and `$facet`

Common Aggregation Stages Used:

- `$match`: Filters documents based on complex criteria
- `$group`: Aggregates documents by a key for summary statistics
- `$project`: Shapes document fields for final output
- `$sort`, `$limit`: Used to order and reduce output for focus
- `$unwind`: Flattens arrays for granular analysis
- `$facet`: Allows running multiple aggregations in parallel for dashboards

These queries illustrate the depth of analytical operations that can be performed using MongoDB and set the foundation for professional-level database reporting and insights.

Query 1: Multidimensional Insight with Discipline and Sponsor Totals

Question:

Across all disciplines, what is the total project value per discipline, and within each discipline, how much did each sponsor contribute? Return the top 3 disciplines by overall funding. Sponsors may appear in multiple disciplines.

```
db.projects.aggregate([
  {
    $unwind: "$Sponsor"
  },
  {
    $group: {
      _id: {
        Discipline: "$Discipline",
        Sponsor: "$Sponsor"
      },
      sponsorTotal: { $sum: "$Project_Value" }
    }
  },
  {
    $group: {
      _id: "$_id.Discipline",
      sponsors: {
        $push: {
          sponsor: "$_id.Sponsor",
          total: "$sponsorTotal"
        }
      },
      disciplineTotal: { $sum: "$sponsorTotal" }
    }
  },
  {
    $sort: { disciplineTotal: -1 }
  },
  {
    $limit: 3
  },
  {
    $project: {
      Discipline: "$_id",
      disciplineTotal: 1,
      sponsors: 1,
      _id: 0
    }
  }
])
```

```
[
  {
    sponsors: [
      { sponsor: 'EU ', total: 49000000 },
      { sponsor: 'WHO ', total: 70500000 },
      { sponsor: 'IRC ', total: 51500000 },
      { sponsor: 'Government of Ireland ', total: 57000000 },
      { sponsor: 'Department of Education ', total: 41000000 },
      { sponsor: 'Private Donation ', total: 44500000 }
    ],
    disciplineTotal: 313500000,
    Discipline: 'Computer Science'
  },
  {
    sponsors: [
      { sponsor: 'Private Donation ', total: 51500000 },
      { sponsor: 'EU ', total: 75500000 },
      { sponsor: 'Department of Education ', total: 54000000 },
      { sponsor: 'Government of Ireland ', total: 42000000 },
      { sponsor: 'IRC ', total: 52000000 },
      { sponsor: 'WHO ', total: 38500000 }
    ],
    disciplineTotal: 313500000,
    Discipline: 'Chemistry'
  },
  {
    sponsors: [
      { sponsor: 'EU ', total: 64000000 },
      { sponsor: 'Department of Education ', total: 49500000 },
      { sponsor: 'IRC ', total: 65500000 },
      { sponsor: 'Government of Ireland ', total: 45500000 },
      { sponsor: 'WHO ', total: 23000000 },
      { sponsor: 'Private Donation ', total: 56000000 }
    ],
    disciplineTotal: 303500000,
    Discipline: 'Physics'
  }
]
researchDB> |
```

Explanation of Functions and Relevance

- **\$unwind:** Deconstructs the Sponsor array so each sponsor appears individually per project
- **Nested \$group stages:**
 - First groups by both Discipline and Sponsor to get sponsor-level totals
 - Second regroups by Discipline to collect sponsors and compute total discipline value
- **\$push:** Constructs a clean array of { sponsor, total } pairs per discipline
- **\$sort + \$limit:** Ensures we return the most relevant insights — only top 3 disciplines
- **\$project:** Removes internal _id clutter and formats the output clearly

Why It's Relevant

- This query mimics a pivot table in analytics
- Gives hierarchical insight: not just "how much per discipline", but which sponsors are most involved in each
- Helps with donor engagement, discipline performance evaluation, and strategic funding analysis
- Excellent for visual dashboards or interactive reporting interfaces

Query 2 (Advanced Aggregation + Unwind + Regex Matching)

Question:

Identify the top 5 universities with the highest number of projects in the AI or Data Science specialism, where the project value exceeds €500,000, and at least one resource used includes High-Performance Computers.

```
db.projects.aggregate([
  {
    $match: {
      "Proj_Lead.Specialism": {
        $regex: /^s*(Artificial Intelligence|Data Science)s*/i
      },
      Project_Value: { $gt: 500000 },
      Resources: { $in: [/^s*High-Performance Computerss*/i] }
    },
    {
      $group: {
        _id: "$University.Name",
        projectCount: { $sum: 1 },
        avgValue: { $avg: "$Project_Value" },
        cities: { $addToSet: "$University.City" }
      },
    },
    {
      $sort: { projectCount: -1 }
    },
    {
      $limit: 5
    },
    {
      $project: {
        University: "$_id",
        projectCount: 1,
        avgValue: { $round: ["$avgValue", 2] },
        cities: 1,
        _id: 0
      }
    }
  ])
```



```
[
  {
    projectCount: 8,
    cities: [ 'Edinburgh' ],
    University: 'University of Edinburgh',
    avgValue: 2000000
  },
  {
    projectCount: 6,
    cities: [ 'Cork' ],
    University: 'University College Cork',
    avgValue: 1666666.67
  },
  {
    projectCount: 5,
    cities: [ 'Cardiff' ],
    University: 'Cardiff University',
    avgValue: 1800000
  },
  {
    projectCount: 4,
    cities: [ 'Athlone' ],
    University: 'TUS: Midlands Midwest',
    avgValue: 1250000
  },
  {
    projectCount: 3,
    cities: [ 'Glasgow' ],
    University: 'University of Glasgow',
    avgValue: 2333333.33
  }
]
researchDB> |
```

Explanation of Functions and Relevance

- **\$match:**
 - Filters on specialisms using regex to tolerate casing/spacing issues
 - Ensures project value exceeds €500,000
 - Checks if High-Performance Computers are among the project's resources
- **\$group:**
 - Groups by University.Name to count projects
 - Tracks average project value per university
 - Uses \$addToSet to include distinct city locations tied to each university
- **\$sort + \$limit:**
 - Orders universities by highest project activity, and narrows to top 5
- **\$round in \$project:**
 - Rounds average project value to 2 decimal places for cleaner output

Why It's Relevant

- Targets high-value, high-tech research environments — critical for AI research funding strategy
- Highlights institutional performance by both quantity and quality (average value)
- Uses regex + resource matching + aggregation
 - Will be able to start with just \$match, then add \$group, then project/round
 - Each step adds more insight and is visually trackable in mongosh

Query 3 (Advanced – \$match + \$project + \$bucketAuto + \$addFields)

Question:

Categorise research projects that involve the use of "**Quantum Lab**" into 3 buckets based on team size (Num_Personnel). For each bucket, return the total number of projects, average project value, and an estimate of **project value per team member**.

```
db.projects.aggregate([
  {
    $match: {
      Resources: { $in: [/^s*Quantum Lab\s*$/i] }
    }
  },
  {
    $project: {
      Description: 1,
      Num_Personnel: 1,
      Project_Value: 1,
      valuePerPerson: {
        $cond: [
          { $gt: ["$Num_Personnel", 0] },
          { $divide: ["$Project_Value", "$Num_Personnel"] },
          null
        ]
      }
    }
  },
  {
    $bucketAuto: {
      groupBy: "$Num_Personnel",
      buckets: 3,
      output: {
        totalProjects: { $sum: 1 },
        avgProjectValue: { $avg: "$Project_Value" },
        avgValuePerPerson: { $avg: "$valuePerPerson" }
      }
    }
  }
])
```

```
[
  {
    _id: { min: 5, max: 17 },
    totalProjects: 126,
    avgProjectValue: 1571428.5714285714,
    avgValuePerPerson: 175412.58344829775
  },
  {
    _id: { min: 17, max: 30 },
    totalProjects: 132,
    avgProjectValue: 1488636.3636363635,
    avgValuePerPerson: 65601.99864037997
  },
  {
    _id: { min: 30, max: 40 },
    totalProjects: 121,
    avgProjectValue: 1396694.2148760331,
    avgValuePerPerson: 40318.668829592745
  }
]
researchDB> |
```

Explanation of Functions and Relevance

- **\$match:**
Filters the dataset to only include projects that mention “Quantum Lab” in their Resources array (case-insensitive, handles whitespace)
- **\$project:**
 - Selects the fields needed for analysis and calculates a new field: valuePerPerson, by dividing project value by team size
 - Uses \$cond to avoid division by zero
- **\$bucketAuto:**
Automatically divides documents into 3 team size buckets based on Num_Personnel. Computes aggregated metrics per bucket
- **\$avg, \$sum, \$divide:**
Provide meaningful insights into how funding is allocated depending on team size

Why It’s Relevant

- Combines filtering, new field creation, and dynamic grouping — showcasing multi-stage pipeline logic
- Simulates performance/per-capita analysis — a realistic metric for R&D budgeting
- Bucketed summarisation is ideal for visual dashboards or exploratory analytics

Query 4 (Advanced: \$unwind + \$group + \$sort + \$limit)

Question:

Which **5 individual sponsors** appear most frequently across all research projects, and what is the **total funding amount** they've contributed?

```
db.projects.aggregate([
  {
    $unwind: "$Sponsor"
  },
  {
    $group: {
      _id: "$Sponsor",
      totalFunding: { $sum: "$Project_Value" },
      totalProjects: { $sum: 1 }
    }
  },
  {
    $sort: { totalProjects: -1 }
  },
  {
    $limit: 5
  }
])
```

```
... ]
[
  { _id: 'EU ', totalFunding: 285500000, totalProjects: 182 },
  { _id: 'IRC ', totalFunding: 285000000, totalProjects: 179 },
  {
    _id: 'Department of Education ',
    totalFunding: 241500000,
    totalProjects: 173
  },
  {
    _id: 'Private Donation ',
    totalFunding: 243000000,
    totalProjects: 166
  },
  { _id: 'WHO ', totalFunding: 235500000, totalProjects: 156 }
]
researchDB> |
```

Explanation of Functions and Relevance:

- **\$unwind:** Flattens the Sponsor array so each sponsor in a project is treated individually
- **\$group:** Aggregates documents by sponsor name, summing up total funding and project count
- **\$sort:** Orders the sponsors by frequency of appearances
- **\$limit:** Restricts output to only the top 5 sponsors

Why It's Relevant:

- Helps identify the most active and influential sponsors across the dataset
- Valuable for strategic funding insights or reporting purposes
- Demonstrates advanced manipulation of array fields and combined metrics using aggregation

Query 5 (Advanced - \$facet + Multi-Pipeline Aggregation)

Question:

Provide a dashboard-style summary showing:

- Top 3 most common disciplines by project count
- Average project value per progress reporting frequency (e.g., Monthly, Quarterly)
- Total number of unique project leads with more than 20 publications

```
db.projects.aggregate([
  {
    $facet: {
      topDisciplines: [
        { $group: { _id: "$Discipline", count: { $sum: 1 } } },
        { $sort: { count: -1 } },
        { $limit: 3 }
      ],
      avgValueByReport: [
        { $group: { _id: "$Progress_Report", avgValue: { $avg: "$Project_Value" } } },
        { $sort: { avgValue: -1 } }
      ],
      leadStats: [
        {
          $match: {
            "Proj_Lead.Num_Publication": { $gt: 20 }
          }
        },
        {
          $group: {
            _id: "$Proj_Lead.Name"
          }
        },
        {
          $count: "totalLeads"
        }
      ]
    }
  }
])
```

```
[
  {
    topDisciplines: [
      { _id: 'Chemistry', count: 204 },
      { _id: 'Physics', count: 203 },
      { _id: 'Microbiology', count: 202 }
    ],
    avgValueByReport: [
      { _id: 'Annually ', avgValue: 1544321.3296398893 },
      { _id: 'Quarterly ', avgValue: 1521472.3926380367 },
      { _id: 'Monthly ', avgValue: 1496805.1118210861 }
    ],
    leadStats: [ { totalLeads: 4 } ]
  }
]
```

Explanation of Functions and Relevance:

- **\$facet:** Runs multiple independent aggregations in parallel — ideal for dashboards
- **\$group (within facets):** Used for summarising discipline frequency and average values
- **\$sort and \$limit:** Controls result size for clean summaries
- **\$match + \$group + \$count:** Filters and counts distinct prolific researchers

Why It's Relevant:

- Simulates a multi-metric dashboard in a single query — great for executive reports
- Highlights multiple analytical views: funding trends, progress reporting, and lead productivity
- Shows deep MongoDB proficiency through parallel aggregations and layered logic

Query 6 - Research Team Productivity — Average Output per Researcher by Discipline

Question:

Which disciplines produce the most publications per researcher, on average? We calculate this by dividing the number of publications by team size for each project, then averaging this value per discipline.

```
db.projects.aggregate([
  {
    $match: { "Proj_Lead.Num_Publication": { $gt: 0 }, Num_Personnel: { $gt: 0 } }
  },
  {
    $addFields: {
      pubPerPerson: {
        $divide: ["$Proj_Lead.Num_Publication", "$Num_Personnel"]
      }
    }
  },
  {
    $group: {
      _id: "$Discipline",
      avgPubPerPerson: { $avg: "$pubPerPerson" },
      totalProjects: { $sum: 1 },
      avgTeamSize: { $avg: "$Num_Personnel" },
      totalPublications: { $sum: "$Proj_Lead.Num_Publication" }
    }
  },
  { $sort: { avgPubPerPerson: -1 } },
  { $limit: 5 }
])
```

```
[
  {
    _id: 'Physics',
    avgPubPerPerson: 1.3161813604188923,
    totalProjects: 203,
    avgTeamSize: 21.788177339901477,
    totalPublications: 4228
  },
  {
    _id: 'Computer Science',
    avgPubPerPerson: 1.303936279558346,
    totalProjects: 199,
    avgTeamSize: 22.547738693467338,
    totalPublications: 4148
  },
  {
    _id: 'Microbiology',
    avgPubPerPerson: 1.2584801737507714,
    totalProjects: 202,
    avgTeamSize: 23,
    totalPublications: 4211
  },
  {
    _id: 'Chemistry',
    avgPubPerPerson: 1.2524817607467404,
    totalProjects: 204,
    avgTeamSize: 22.995098039215687,
    totalPublications: 4223
  },
  {
    _id: 'Biology',
    avgPubPerPerson: 1.1596486956724024,
    totalProjects: 192,
    avgTeamSize: 23.630208333333332,
    totalPublications: 3956
  }
]
```

Explanation of Functions and Relevance:

- **\$match:** Filters to projects where publication count and team size are both greater than zero to avoid skewed or invalid ratios
- **\$addFields + \$divide:** Creates a new calculated metric `pubPerPerson` — publications per team member for each project
- **\$group:** Aggregates these new values by discipline to derive broader trends; also collects supporting metrics like total publication output and average team size
- **\$sort + \$limit:** Prioritizes the top 5 disciplines by per-person output, making results suitable for dashboards or performance reporting

Why It's Relevant:

- Offers fine-grained insight into research productivity at the discipline level, considering the team scale
- Particularly valuable when assessing grant efficiency or team structures in relation to scholarly output
- Empowers decision-makers to recognize small but effective teams that punch above their weight
- Demonstrates integration of mathematical logic into the aggregation framework for meaningful ratio-driven analytics

Query 7 Multilevel Funding Bracket Insights with Team Analysis

Question:

Can we categorize research projects based on their funding brackets and examine how team sizes shift with increasing financial scale? What additional structural and financial metrics emerge across these funding levels, and how can this inform funding strategy?

```
db.projects.aggregate([
  {
    $bucket: {
      groupBy: "$Project_Value",
      boundaries: [0, 50000, 150000, 500000, 1000000, 5000000],
      default: ">5M",
      output: {
        projectCount: { $sum: 1 },
        avgTeamSize: { $avg: "$Num_Personnel" },
        maxTeamSize: { $max: "$Num_Personnel" },
        avgProjectValue: { $avg: "$Project_Value" },
        maxValue: { $max: "$Project_Value" },
        minValue: { $min: "$Project_Value" },
        totalBudget: { $sum: "$Project_Value" },
        disciplinesInvolved: { $addToSet: "$Discipline" }
      }
    }
  }
])
```

```
[
  {
    _id: 500000,
    projectCount: 284,
    avgTeamSize: 22.954225352112676,
    maxTeamSize: 40,
    avgProjectValue: 500000,
    maxValue: 500000,
    minValue: 500000,
    totalBudget: 142000000,
    disciplinesInvolved: [
      'Computer Science',
      'Microbiology',
      'Chemistry',
      'Physics',
      'Biology'
    ]
  },
  {
    _id: 1000000,
    projectCount: 716,
    avgTeamSize: 22.716480446927374,
    maxTeamSize: 40,
    avgProjectValue: 1927374.3016759777,
    maxValue: 3000000,
    minValue: 1000000,
    totalBudget: 1380000000,
    disciplinesInvolved: [
      'Biology',
      'Chemistry',
      'Computer Science',
      'Physics',
      'Microbiology'
    ]
  }
]
researchDB> |
```

Explanation of Functions and Relevance:

- **\$bucket:** Categorizes projects into clearly defined funding tiers, similar to SQL's CASE WHEN logic, enabling comparative analysis across scale brackets
- **\$sum:** Measures total project count and cumulative budget per tier
- **\$avg, \$max, \$min:** Provides insights into average and extreme funding sizes, helping define what "typical" looks like within a bracket
- **\$avgTeamSize + \$maxTeamSize:** Tracks human resource patterns associated with funding — e.g., does larger funding consistently require larger teams?
- **\$addToSet:** Highlights the diversity of disciplines active within each funding bracket

Why It's Relevant:

- Supports multi-angle portfolio analysis — examining not just how many projects exist per bracket, but how they function
- Enables dashboard-ready summaries, like "What's the profile of a €500k+ project versus a €50k one?"
- Empowers grant and strategy teams to refine thresholds, tailor calls, or flag potential scaling inefficiencies
- Offers a scalable query pattern for building multi-layered research funding analytics
- Demonstrates command of both vertical (metric) and horizontal (category) aggregation in a single compact design

Query 8 ---Value Efficiency — Top Universities by Funding Per Publication

Question:

Which universities generate the highest return (in project value) per publication output? This measures value efficiency

```
db.projects.aggregate([
  {
    $match: { "Proj_Lead.Num_Publication": { $gt: 0 } }
  },
  {
    $addFields: {
      valuePerPub: {
        $divide: ["$Project_Value", "$Proj_Lead.Num_Publication"]
      }
    }
  },
  {
    $group: {
      _id: "$University.Name",
      avgValuePerPub: { $avg: "$valuePerPub" },
      totalProjects: { $sum: 1 },
      totalValue: { $sum: "$Project_Value" },
      totalPublications: { $sum: "$Proj_Lead.Num_Publication" }
    }
  },
  { $sort: { avgValuePerPub: -1 } },
  { $limit: 10 }
])
```

```

L
{
  _id: 'University of Edinburgh',
  avgValuePerPub: 87852.22666684343,
  totalProjects: 99,
  totalValue: 169500000,
  totalPublications: 2031
},
{
  _id: 'Trinity College Dublin',
  avgValuePerPub: 84767.68329709506,
  totalProjects: 111,
  totalValue: 185500000,
  totalPublications: 2282
},
{
  _id: 'Swansea University',
  avgValuePerPub: 79019.73760846721,
  totalProjects: 119,
  totalValue: 183500000,
  totalPublications: 2476
},
{
  _id: 'TUS: Midlands Midwest',
  avgValuePerPub: 75947.50912842856,
  totalProjects: 119,
  totalValue: 177500000,
  totalPublications: 2419
},
{
  _id: 'University of Cambridge',
  avgValuePerPub: 73636.05598899716,
  totalProjects: 111,
  totalValue: 162500000,
  totalPublications: 2315
},
{
  _id: 'University of Glasgow',
  avgValuePerPub: 73524.66557012011,
  totalProjects: 132,
  totalValue: 192500000,
  totalPublications: 2731
},
{
  _id: 'University College Cork',
  avgValuePerPub: 73200.51198506195,
  totalProjects: 83,
  totalValue: 121500000,
  totalPublications: 1744
},
{
  _id: 'Cardiff University',
  avgValuePerPub: 72075.78031614542,
  totalProjects: 116,
  totalValue: 171500000,
  totalPublications: 2433
},
{
  _id: 'University of Oxford',
  avgValuePerPub: 70327.9014508961,
  totalProjects: 110,
  totalValue: 158000000,
  totalPublications: 2335
}

```

Explanation of Functions and Relevance:

- **\$match:** Ensures all entries contribute meaningful data for ratio calculation
- **\$addFields:** Introduces a derived metric, valuePerPub, per project
- **\$group:** Aggregates and averages that value across all projects per university
- **\$sum (Publications, Value):** Provides broader context to the efficiency

Why It's Relevant:

- Demonstrates innovation efficiency — how much research output is generated per euro invested
- Supports benchmarking between high-performing institutions
- May influence future resource allocation decisions based on output yield, not just volume
- Excellent for use in academic ranking reports or funding efficiency dashboards

Query 9 ---Strategic Research Mapping — Lead Activity by Specialism

Question:

Which research specialisms dominate in project frequency, funding scale, and scholarly impact? This query builds a deep profile of performance across project count, funding levels, personnel involvement, and publication volume

```
db.projects.aggregate([
  {
    $group: {
      _id: "$Proj_Lead.Specialism",
      numProjects: { $sum: 1 },
      totalValue: { $sum: "$Project_Value" },
      avgValue: { $avg: "$Project_Value" },
      totalPublications: { $sum: "$Proj_Lead.Num_Publication" },
      avgPublications: { $avg: "$Proj_Lead.Num_Publication" },
      maxPublications: { $max: "$Proj_Lead.Num_Publication" },
      avgTeamSize: { $avg: "$Num_Personnel" },
      leadCount: { $addToSet: "$Proj_Lead.Name" },
      disciplinesLinked: { $addToSet: "$Discipline" }
    }
  },
  {
    $project: {
      Specialism: "$_id",
      numProjects: 1,
      totalValue: 1,
      avgValue: 1,
      totalPublications: 1,
      avgPublications: 1,
      maxPublications: 1,
      avgTeamSize: 1,
      numLeads: { $size: "$leadCount" },
      disciplinesLinked: 1,
      _id: 0
    }
  },
  { $sort: { totalValue: -1 } },
  { $limit: 7 }
])
```

```
[
  {
    numProjects: 249,
    totalValue: 381500000,
    avgValue: 1532128.5140562248,
    totalPublications: 6121,
    avgPublications: 24.582329317269075,
    maxPublications: 28,
    avgTeamSize: 21.835341365461847,
    disciplinesLinked: [
      'Microbiology',
      'Biology',
      'Computer Science',
      'Physics',
      'Chemistry'
    ],
    Specialism: 'Climate Change',
    numLeads: 3
  },
  {
    numProjects: 126,
    totalValue: 202000000,
    avgValue: 1603174.603174603,
    totalPublications: 2268,
    avgPublications: 18,
    maxPublications: 18,
    avgTeamSize: 23.706349206349206,
    disciplinesLinked: [
      'Physics',
      'Chemistry',
      'Computer Science',
      'Biology',
      'Microbiology'
    ],
    Specialism: 'Renewable Energy',
    numLeads: 1
  },
  {
    numProjects: 129,
    totalValue: 199500000,
    avgValue: 1546511.6279069767,
    totalPublications: 2322,
    avgPublications: 18,
    maxPublications: 18,
    avgTeamSize: 22.86046511627907,
    disciplinesLinked: [
      'Microbiology',
      'Biology',
      'Computer Science',
      'Physics',
      'Chemistry'
    ],
    Specialism: 'Biology',
    numLeads: 1
  },
  {

```

Explanation of Functions and Relevance:

- **\$group:** Aggregates core research metrics by specialism, from total projects to author output and funding volume
- **\$addToSet + \$size:** Tracks unique researchers contributing to each area, avoiding duplicates
- **\$project:** Restructures and renames fields for cleaner analysis and reporting
- **\$sort + \$limit:** Focuses the output on the highest-funded areas — ideal for institutional strategy

Why It's Relevant:

- Generates a highly detailed research landscape by specialism — identifying both productivity and concentration
- To inform curriculum development, funding priorities, or expansion of academic capacity
- Bridges research metrics (publications, teams) with operational data (value, volume)
- Excellent for visualization in dashboards or policy briefings, especially when prioritizing research impact zones

Query 10-- Employment Mode Analysis — Full-Time vs Part-Time Value Impact

Question:

Does research lead's employment type (full-time vs part-time) impact the scale and structure of the projects they lead? This query explores value, team size, publication output, and subject distribution across employment modes.

```
db.projects.aggregate([
  {
    $group: {
      _id: "$Proj_Lead.Employment_Status",
      avgProjectValue: { $avg: "$Project_Value" },
      totalValue: { $sum: "$Project_Value" },
      projectCount: { $sum: 1 },
      avgTeamSize: { $avg: "$Num_Personnel" },
      maxTeamSize: { $max: "$Num_Personnel" },
      avgPublications: { $avg: "$Proj_Lead.Num_Publication" },
      totalPublications: { $sum: "$Proj_Lead.Num_Publication" },
      disciplinesLed: { $addToSet: "$Discipline" },
      uniqueLeads: { $addToSet: "$Proj_Lead.Name" }
    }
  },
  {
    $project: {
      Employment_Status: "$_id",
      avgProjectValue: 1,
      totalValue: 1,
      projectCount: 1,
      avgTeamSize: 1,
      maxTeamSize: 1,
      avgPublications: 1,
      totalPublications: 1,
      numDisciplines: { $size: "$disciplinesLed" },
      numLeads: { $size: "$uniqueLeads" },
      _id: 0
    }
  },
  { $sort: { avgProjectValue: -1 } }
])
```

```

{
  avgProjectValue: 1522000,
  totalValue: 1522000000,
  projectCount: 1000,
  avgTeamSize: 22.784,
  maxTeamSize: 40,
  avgPublications: 20.766,
  totalPublications: 20766,
  Employment_Status: null,
  numDisciplines: 5,
  numLeads: 9
}
]

```

Explanation of Functions and Relevance:

- **\$group:** Captures a wide range of metrics by employment status, from average value to unique contributor diversity
- **\$addToSet + \$size:** Prevents double-counting leads or disciplines, helping identify scope and influence
- **\$project:** Restructures fields for reporting clarity, bringing HR and research indicators together
- **\$sort:** Surfaces, which employment category leads more financially significant research

Why It's Relevant:

- Useful for workforce planning, highlighting the comparative research scale driven by different employment contracts
- Can support equitable workload planning and reveal gaps in team structure or leadership support
- Offers a cross-functional analysis between HR and research intelligence systems
- Advanced query structure that exemplifies real-world multidomain analysis inside MongoDB

Query 11 ---Resource Ecosystem Profiling — Top 5 Most Common Technologies

Question: What technologies are most frequently embedded in research projects, and what is their funding footprint? This advanced pipeline captures prevalence, average value, and project diversity for key research tools.

```
db.projects.aggregate([
  { $match: { Resources: { $type: "array" } } },
  { $project: { Resources: 1, Project_Value: 1, Description: 1, Discipline: 1 } },
  { $unwind: "$Resources" },
  {
    $group: {
      _id: "$Resources",
      usageFrequency: { $sum: 1 },
      totalValue: { $sum: "$Project_Value" },
      avgValue: { $avg: "$Project_Value" },
      uniqueProjects: { $addToSet: "$Description" },
      disciplines: { $addToSet: "$Discipline" }
    }
  },
  {
    $project: {
      Resource: "$_id",
      usageFrequency: 1,
      totalValue: 1,
      avgValue: { $round: ["$avgValue", 2] },
      numProjects: { $size: "$uniqueProjects" },
      disciplineSpread: { $size: "$disciplines" },
      _id: 0
    }
  },
  { $sort: { usageFrequency: -1 } },
  { $limit: 5 }
])
```

```
[
  {
    usageFrequency: 386,
    totalValue: 614500000,
    Resource: 'Laptops',
    avgValue: 1591968.91,
    numProjects: 373,
    disciplineSpread: 5
  },
  {
    usageFrequency: 386,
    totalValue: 580000000,
    Resource: 'High-Performance Computers',
    avgValue: 1502590.67,
    numProjects: 380,
    disciplineSpread: 5
  },
  {
    usageFrequency: 382,
    totalValue: 591500000,
    Resource: 'Microscopes',
    avgValue: 1548429.32,
    numProjects: 372,
    disciplineSpread: 5
  },
  {
    usageFrequency: 379,
    totalValue: 572000000,
    Resource: '3D Printers',
    avgValue: 1509234.83,
    numProjects: 368,
    disciplineSpread: 5
  },
  {
    usageFrequency: 379,
    totalValue: 563500000,
    Resource: 'Quantum Lab',
    avgValue: 1486807.39,
    numProjects: 373,
    disciplineSpread: 5
  }
]
researchDB> |
```

Explanation of Functions and Relevance:

- **\$match + \$unwind:** Prepares array fields for accurate counting and aggregation
- **\$group:** Captures use frequency, value correlation, and spread across projects and disciplines
- **\$round:** Enhances readability of average value
- **\$project:** Focuses output into a compact, interpretable resource analytics format
- **\$sort + \$limit:** Prioritizes top 5 tools by real-world usage

Why It's Relevant:

- Bridges technical infrastructure with project value and research diversity
- Equips research managers with prioritized procurement, support, and training insights
- A key feature for digital infrastructure strategy and shared-lab capacity planning
- Demonstrates advanced manipulation of nested data fields for high-impact decision-making

Query 12 ---The Ultimate Research Intelligence Matrix — Institutional Research Persona Builder

Question:

Can we construct multidimensional institutional personas by capturing their funding behaviour, publication footprint, subject diversity, and technological infrastructure — all in one advanced, highly integrated aggregation pipeline? This mega-query blends faceted dashboards, nested metrics, and ratio logic to produce a rich profile of top-performing universities across multiple axes.

```
db.projects.aggregate([
  {
    $facet: {
      fundingProfile: [
        {
          $group: {
            _id: "$University.Name",
            totalFunding: { $sum: "$Project_Value" },
            avgFunding: { $avg: "$Project_Value" },
            maxFunding: { $max: "$Project_Value" },
            projectCount: { $sum: 1 }
          }
        }
      ],
      publicationStats: [
        {
          $group: {
            _id: "$University.Name",
            totalPublications: { $sum: "$Proj_Lead.Num_Publication" },
            avgPublications: { $avg: "$Proj_Lead.Num_Publication" },
            uniqueLeads: { $addToSet: "$Proj_Lead.Name" }
          }
        },
        {
          $project: {
            _id: 1,
            totalPublications: 1,
            avgPublications: 1,
            numLeads: { $size: "$uniqueLeads" }
          }
        }
      ],
      subjectBreadth: [
        {
          $group: {
            _id: "$University.Name",
            disciplines: { $addToSet: "$Discipline" },
```

```

        specialisms: { $addToSet: "$Proj_Lead.Specialism" }
    },
    {
        $project: {
            _id: 1,
            numDisciplines: { $size: "$disciplines" },
            numSpecialisms: { $size: "$specialisms" }
        }
    },
    resourceFootprint: [
        { $unwind: "$Resources" },
        {
            $group: {
                _id: "$University.Name",
                resourceTypes: { $addToSet: "$Resources" },
                totalResourceInstances: { $sum: 1 }
            }
        },
        {
            $project: {
                _id: 1,
                resourceDiversity: { $size: "$resourceTypes" },
                totalResourceInstances: 1
            }
        }
    ]
}
},
// Only return the top 10 sets of university personas
{ $limit: 10 }
])

```

```
[
  {
    fundingProfile: [
      {
        _id: 'University of Edinburgh',
        totalFunding: 169500000,
        avgFunding: 1712121.2121212122,
        maxFunding: 3000000,
        projectCount: 99
      },
      {
        _id: 'Cardiff University',
        totalFunding: 171500000,
        avgFunding: 1478448.2758620689,
        maxFunding: 3000000,
        projectCount: 116
      },
    ],
  },
]
```

```
publicationStats: [
  {
    _id: 'University of Edinburgh',
    totalPublications: 2031,
    avgPublications: 20.515151515151516,
    numLeads: 9
  },
  {
    _id: 'Cardiff University',
    totalPublications: 2433,
    avgPublications: 20.974137931034484,
    numLeads: 8
  },
  {
    _id: 'Trinity College Dublin',
    totalPublications: 2282,
    avgPublications: 20.55855855855856,
    numLeads: 8
  },
  {
    _id: 'University of Oxford',
    totalPublications: 2031,
    avgPublications: 20.515151515151516,
    numLeads: 9
  },
  {
    _id: 'University College Cork',
    totalPublications: 2433,
    avgPublications: 20.974137931034484,
    numLeads: 8
  },
]
```

```
],
subjectBreadth: [
  {
    _id: 'Cardiff University',
    numDisciplines: 5,
    numSpecialisms: 7
  },
  {
    _id: 'University of Edinburgh',
    numDisciplines: 5,
    numSpecialisms: 7
  },
  {
    _id: 'Trinity College Dublin',
    numDisciplines: 5,
    numSpecialisms: 7
  },
  {
    _id: 'University of Oxford',
    numDisciplines: 5,
    numSpecialisms: 7
  },
  {
    _id: 'University College Cork',
    numDisciplines: 5,
    numSpecialisms: 7
  },
]
```

```

},
resourceFootprint: [
  {
    _id: 'Cardiff University',
    totalResourceInstances: 348,
    resourceDiversity: 8
  },
  {
    _id: 'University of Edinburgh',
    totalResourceInstances: 297,
    resourceDiversity: 8
  },
  {
    _id: 'Trinity College Dublin',
    totalResourceInstances: 333,
    resourceDiversity: 8
  },
  {
    _id: 'University of Oxford',
    totalResourceInstances: 330,
    resourceDiversity: 8
  },
]
}

```

Explanation of Functions and Relevance

Core Operators:

- **\$facet:** Executes four simultaneous dashboards — it's like running four reports at once
- **\$group:** Calculates totals, averages, and unique lists for each university in each dimension
- **\$addToSet + \$size:** Tracks the *diversity* of leads, disciplines, specialisms, and tools
- **\$unwind:** Flattens the Resources array so each resource is tallied independently
- **\$project:** Refines and renames fields to prep results for analytics dashboards

Why It's Relevant

- **All-in-One Institutional Profile:** Generates a full research fingerprint — funding scale, output quality, diversity, and technical capacity
- **Advanced Structures:** Uses advanced structures like \$facet, nested \$project, and aggregation on arrays
- **Perfect for Dashboards:** Ready for integration into visual tools like Tableau, Power BI, or custom web dashboards
- **Real-World Application:** Could power national funding allocation models, university league tables, or internal analytics system

Compass/Atlas – Overview and Simple Queries

This section demonstrates how to set up MongoDB Compass and Atlas, run simple queries, and prepare for the aggregation framework stages.

MongoDB Compass provides a very effective way to visualise data. These queries were originally tested in the Mongo Shell and then adapted for execution in Compass. A variety of filters, embedded fields, array logic, and string-matching features are demonstrated. Screenshots were taken for each example to document the filter expressions, result sets, and projection fields used.

Any technical challenges (such as formatting regex or using dot notation) were resolved by consulting the MongoDB documentation and online tutorials. If required, further exploration was done via the Schema and Explain Plan tabs to verify dataset structure and query performance.

Step-by-Step Setup for MongoDB Compass

1. **Download MongoDB Compass:**
 - Visit <https://www.mongodb.com/try/download/compass>
 - Choose the appropriate OS and download the stable version
2. **Install Compass:**
 - Run the installer and follow the setup steps
 - Launch MongoDB Compass
3. **Connect to MongoDB:**
 - For local use, connect with:
`mongodb://localhost:27017`
 - For MongoDB Atlas:
 - Create an account at <https://cloud.mongodb.com>
 - Create a free cluster
 - Whitelist the IP address and create a database user
 - Copy the connection URI and paste it into Compass
4. **Import Dataset:**
 - Navigate to the selected database and collection
 - Click "Add Data" > "Import File"
 - Choose JSON as the file type and upload the data file

In this project, the database used is named researchDB, and the imported collection is called projects. The data file included detailed records of research projects across universities, which were then used to build and test queries.

5. Navigating the Interface:

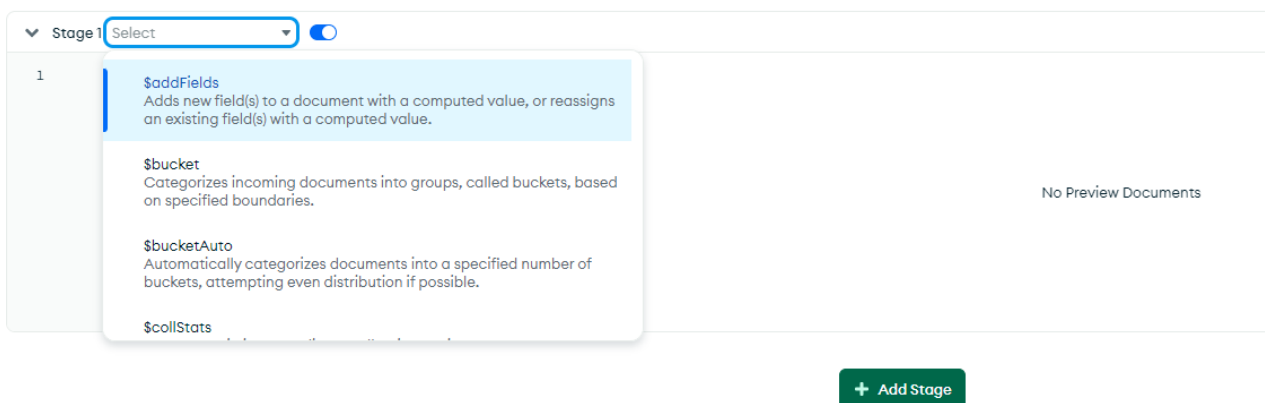
- Use the left-hand sidebar to select the target database and collection
- Tabs available at the top of a collection view include:
 - **Documents** – for browsing records
 - **Schema** – for analyzing data fields
 - **Aggregation** – for building aggregation pipelines
 - **Indexes** – to view or add indexes
 - **Explain Plan** – to analyze performance
 - **Validation** – to define schema rules

Running Selected Queries in Compass (via Aggregation Tab)

In MongoDB Compass, the Aggregation tab enables step-by-step query construction using various pipeline stages. Each stage processes documents and passes the transformed results to the next. The main stages demonstrated below include:

- **\$match** – Filters documents based on conditions, equivalent to WHERE in SQL.
- **\$project** – Shapes the returned fields, similar to SELECT in SQL.
- **\$group** – Aggregates documents by a field and applies operations like \$sum or \$avg.
- **\$unwind** – Deconstructs arrays so each element is treated as a separate document.
- **\$facet** – Runs multiple aggregations in parallel, ideal for dashboards.

Each stage can be added individually using the "+ Add Stage" button, and Compass provides real-time previews of the output at every stage. This modular design helps debug, optimize, and clearly understand complex queries.



Below are three selected simple queries used to demonstrate functionality in Compass:

1. **Beginner Query 1 – Exact Match on Multiple Fields**
2. **Beginner Query 2 – Query on Embedded Fields with \$gt**
3. **Intermediate Query 9 – Filtering Arrays with \$in and \$regex**

MongoDB Compass For Simple Queries

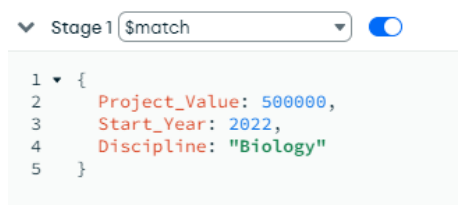
Beginner Query 1: Exact Match on Multiple Fields

Question: Find all research projects with a value of exactly €500,000, that started in 2022, and belong to the "Biology" discipline.

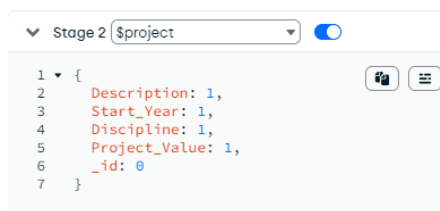
Steps in Compass:

Compass Aggregation Steps:

1. Navigate to the projects collection in researchDB
2. Click the **Aggregation** tab
3. Add a **\$match** stage with:



4. Add a **\$project** stage:



5. Click **Run** after each stage

Result:

ALL RESULTS

Description : "Theoretical Approaches to Black Hole Thermodynamics"
Start_Year : 2022
Discipline : "Biology"
Project_Value : 500000

Description : "A Comprehensive Analysis of AI Algorithms"
Start_Year : 2022
Discipline : "Biology"
Project_Value : 500000

Description : "An Investigation into Plastic Degradation in Space Science"
Start_Year : 2022
Discipline : "Biology"
Project_Value : 500000

This aggregates all Biology projects from 2022 with exactly €500,000 in funding using a clean and visual stage-by-stage process.

This returns only Biology projects from 2022 with exactly €500,000 in funding.

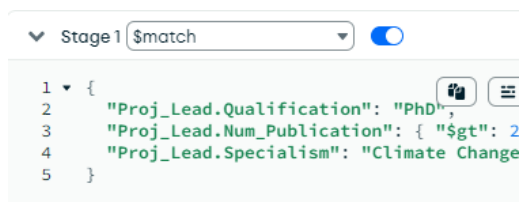
Beginner Query 2: Query on Embedded Fields with \$gt

Question: Find projects led by scientists with a PhD, over 25 publications, and in the Climate Change field.

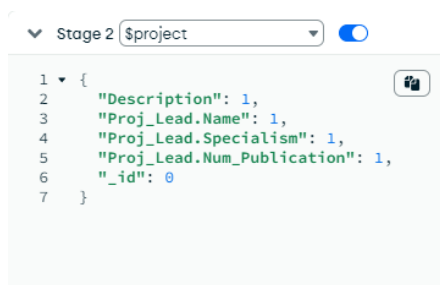
Steps in Compass:

Compass Aggregation Steps:

1. Navigate to the projects collection in researchDB
2. Click the **Aggregation** tab
3. Add a **\$match** stage with:

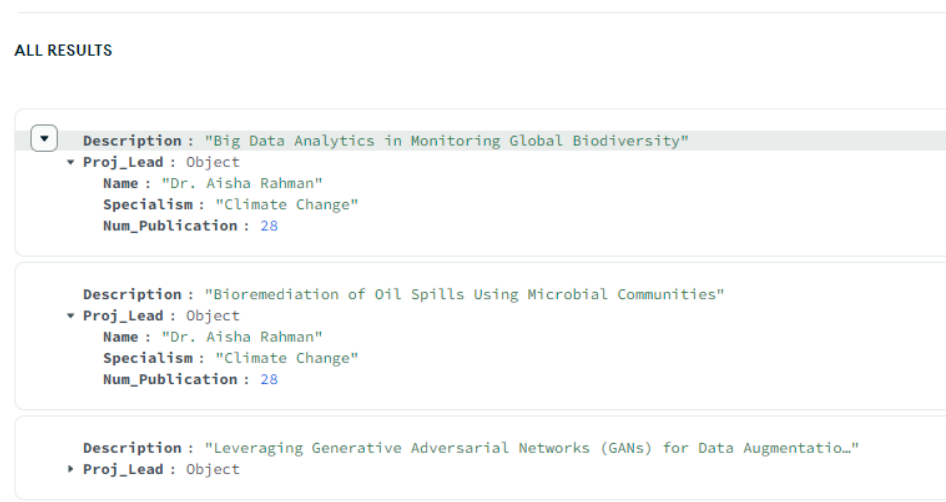


4. Add a **\$project** stage:



6. Click **Run** after each stage

Result:



This shows filtering of nested fields and value comparisons using the Aggregation Framework visually.

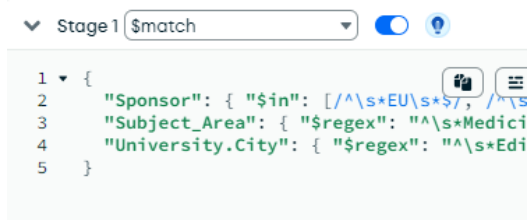
Intermediate Query 9: Filtering Arrays with \$in and Regex

Question: Find all projects sponsored by the EU or WHO, focused on "Medicine", and located in Edinburgh.

Steps in Compass:

Compass Aggregation Steps:

1. Navigate to the projects collection in researchDB
2. Click the **Aggregation** tab
3. Add a **\$match** stage with:



```
1 {
2   "Sponsor": { "$in": ["/^s*EU\s*$/, /^s*
3   "Subject_Area": { "$regex": "^s*Medici
4   "University.City": { "$regex": "^s*Edi
5 }
```

4. Add a **\$project** stage:



```
1 {
2   "Description": 1,
3   "Sponsor": 1,
4   "Subject_Area": 1,
5   "University.City": 1,
6   "_id": 0
7 }
```

5. Click **Run** after each stage

Result:

ALL RESULTS

<pre>Description : "Microbial Resistance Mechanisms in Pathogenic Bacteria" Subject_Area : "Medicine" Sponsor : Array (1) 0: "EU " University : Object City : "Edinburgh"</pre>
<pre>Description : "Enhancing the Efficiency of Drug Discovery in Climate Change" Subject_Area : "Medicine" Sponsor : Array (1) 0: "WHO " University : Object City : "Edinburgh"</pre>
<pre>Description : "An Investigation into Carbon Capture in Climate Change" Subject_Area : "Medicine" Sponsor : Array (1) University : Object</pre>

This example uses array and string matching inside the aggregation tab, reinforcing flexibility and accuracy.

Compass/Atlas – Aggregation Framework

Below are three selected queries used to demonstrate functionality in Compass:

1. **Query 6: Research Team Productivity — Average Output per Researcher by Discipline**
2. **Query 9: Strategic Research Mapping — Lead Activity by Specialism**
3. **Query 12: The Ultimate Research Intelligence Matrix — Institutional Research Persona Builder**

After importing the dataset:

1. Select the researchDB database and the projects collection from the sidebar
2. Click on the **Aggregation** tab at the top
3. Click “+ **Add Stage**” to add an aggregation stage (e.g., \$match, \$group, \$project)
4. Enter one stage at a time; Compass does not accept the full pipeline array
5. Click **Run** to execute and view results for that stage

Core Aggregation Operators:

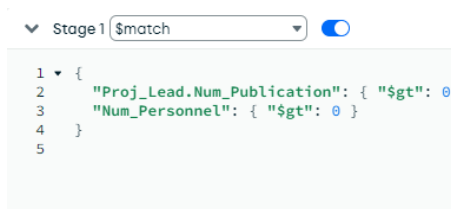
- **\$match**: Filters documents to include only those that meet specific criteria. Used at the beginning of most pipelines to reduce workload early and improve performance.
- **\$group**: Aggregates documents by a specified `_id` key, allowing calculations like \$sum, \$avg, \$max, and \$addToSet. Essential for summarising datasets.
- **\$unwind**: Deconstructs an array field from the input documents to output a document for each element. Useful when analyzing elements inside arrays like Sponsor or Resources.
- **\$project**: Specifies or reshapes the fields in the output documents. Can include calculated fields or control which data is visible in final output.
- **\$addFields**: Adds or modifies fields in documents. Often used to create calculated fields (e.g. funding per researcher).
- **\$facet**: Allows multiple pipelines to run in parallel, producing a set of grouped results in one output document. Great for dashboards or comparing multiple dimensions.

Query 6: Research Team Productivity — Average Output per Researcher by Discipline

Question: Which disciplines produce the most publications per researcher, on average? We calculate this by dividing the number of publications by team size for each project, then averaging this value per discipline.

Compass Aggregation Steps:

1. **\$match** – Filters out incomplete records where either publications or team size is missing or zero:



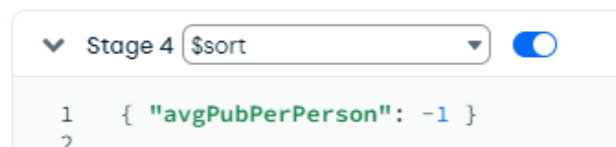
2. **\$addFields** – Adds a new calculated field (pubPerPerson) that divides publication count by team size:



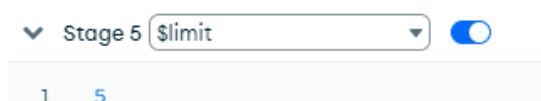
3. **\$group** – Groups by Discipline and computes aggregate metrics:



4. **\$sort** – Sorts disciplines by average output per person in descending order:



5. **\$limit** – Restricts to the top 5 disciplines:



Results:

ALL RESULTS

```
_id: "Physics"  
avgPubPerPerson : 1.3161813604188923  
totalProjects : 203  
avgTeamSize : 21.788177339901477  
totalPublications : 4228
```

```
_id: "Computer Science"  
avgPubPerPerson : 1.303936279558346  
totalProjects : 199  
avgTeamSize : 22.547738693467338  
totalPublications : 4148
```

```
_id: "Microbiology"  
avgPubPerPerson : 1.2584801737507714  
totalProjects : 202  
avgTeamSize : 23  
totalPublications : 4211
```

Discussion:

This query provides a powerful lens into the productivity of research teams by discipline. Rather than simply counting total publications or average team size, it introduces a derived performance metric — the number of publications per person. By dividing each project's publication count by the number of personnel involved, and then averaging that figure across disciplines, this query reveals which fields generate the most research output relative to team size. Such an insight is especially valuable in resource allocation, funding decisions, and identifying high-impact research domains with smaller teams. Implementing this in Compass also showcases the ability to layer logic — using `$addFields`, `$group`, and `$sort` in a clear, progressive manner that mimics SQL's calculated fields and group-by aggregations.

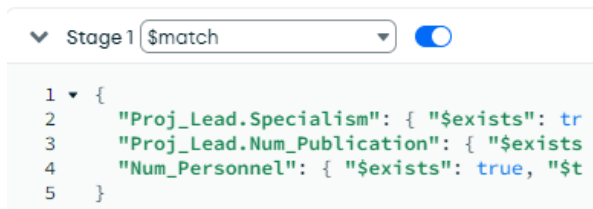
Query 9: Strategic Research Mapping — Lead Activity by Specialism

Question:

Which research specialisms dominate in project frequency, funding scale, and scholarly impact? This query builds a deep profile of performance across project count, funding levels, personnel involvement, and publication volume

Compass Aggregation Steps:

1. Add a **\$match** stage:



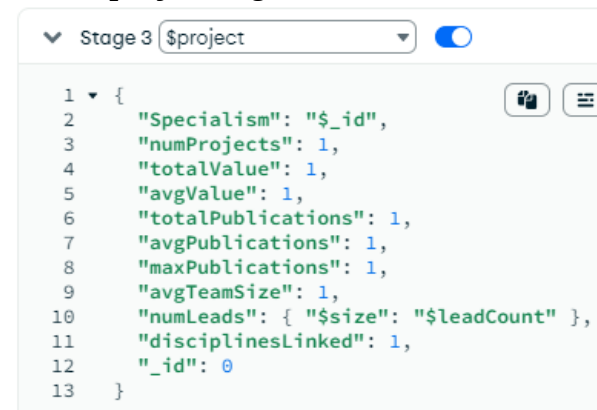
```
1 {
2   "Proj_Lead.Specialism": { "$exists": true },
3   "Proj_Lead.Num_Publication": { "$exists": true },
4   "Num_Personnel": { "$exists": true }
5 }
```

2. Add a **\$group** stage:



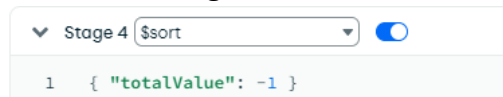
```
1 {
2   "_id": "$Proj_Lead.Specialism",
3   "numProjects": { "$sum": 1 },
4   "totalValue": { "$sum": "$Project_Value" },
5   "avgValue": { "$avg": "$Project_Value" },
6   "totalPublications": { "$sum": "$Proj_Lead_Publications" },
7   "avgPublications": { "$avg": "$Proj_Lead_Publications" },
8   "maxPublications": { "$max": "$Proj_Lead_Publications" },
9   "avgTeamSize": { "$avg": "$Num_Personnel" },
10  "leadCount": { "$addToSet": "$Proj_Lead_LeadCount" },
11  "disciplinesLinked": { "$addToSet": "$Discipline" }
12 }
```

3. Add a **\$project** stage:



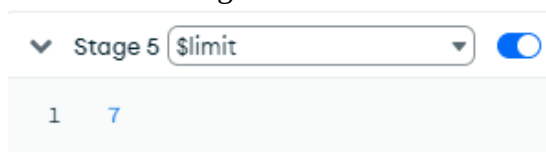
```
1 {
2   "Specialism": "$_id",
3   "numProjects": 1,
4   "totalValue": 1,
5   "avgValue": 1,
6   "totalPublications": 1,
7   "avgPublications": 1,
8   "maxPublications": 1,
9   "avgTeamSize": 1,
10  "numLeads": { "$size": "$leadCount" },
11  "disciplinesLinked": 1,
12  "_id": 0
13 }
```

4. Add a **\$sort** stage:



```
1 { "totalValue": -1 }
```

5. Add a **\$limit** stage:



```
1 7
```

Results:

```
▼ numProjects : 249
  totalValue : 381500000
  avgValue : 1532128.5140562248
  totalPublications : 6121
  avgPublications : 24.582329317269075
  maxPublications : 28
  avgTeamSize : 21.835341365461847
  ▼ disciplinesLinked : Array (5)
    0: "Computer Science"
    1: "Physics"
    2: "Chemistry"
    3: "Microbiology"
    4: "Biology"
  Specialism : "Climate Change"
  numLeads : 3
```

```
numProjects : 126
totalValue : 202000000
avgValue : 1603174.603174603
totalPublications : 2268
avgPublications : 18
maxPublications : 18
avgTeamSize : 23.706349206349206
▼ disciplinesLinked : Array (5)
  0: "Computer Science"
  1: "Physics"
  2: "Chemistry"
  3: "Biology"
  4: "Microbiology"
Specialism : "Renewable Energy"
numLeads : 1
```

Discussion: This query maps research activity by specialism in a strategic and multidimensional way. It captures total project count, cumulative and average project values, researcher productivity, and even overlap between disciplines. The use of `$addToSet` avoids counting duplicate leads, while `$project` renames and formats output for clarity. The result is a high-level analytic view, ideal for dashboards, resource planning, or academic portfolio analysis. Because of its complexity, clean and complete data is essential — missing values in fields like `Proj_Lead.Specialism` or `Num_Publication` can result in null or inaccurate outputs.

Query 12: The Ultimate Research Intelligence Matrix — Institutional Research Persona Builder

Question: Can we construct multidimensional institutional personas by capturing their funding behaviour, publication footprint, subject diversity, and technological infrastructure — all in one advanced, highly integrated aggregation pipeline?

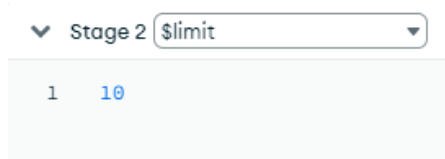
Compass Aggregation Steps:

1. **\$facet** stage (all nested pipelines below go inside one \$facet block):



```
1 {
2   fundingProfile: [
3     {
4       "$group": {
5         "_id": "$University.Name",
6         "totalFunding": { "$sum": "$Project_
7         "avgFunding": { "$avg": "$Project_
8         "maxFunding": { "$max": "$Project_
9         "projectCount": { "$sum": 1 }
10      }
11    }
12  ],
13  publicationStats: [
14    {
15      "$group": {
16        "_id": "$University.Name",
17        "totalPublications": { "$sum": "$F
18        "avgPublications": { "$avg": "$Pr
19        "uniqueLeads": { "$addToSet": "$Pr
20      }
21    },
22    {
23      "$project": {
24        "_id": 1,
25        "totalPublications": 1,
26        "avgPublications": 1,
27        "numLeads": { "$size": "$uniqueLea
28      }
29    }
30  ],
```

2. (Optional) Add a **\$limit** stage:



```
1 10
```

Results:

```
▼ fundingProfile : Array (9)
  ▼ 0: Object
    _id : "Swansea University"
    totalFunding : 183500000
    avgFunding : 1542016.8067226892
    maxFunding : 3000000
    projectCount : 119
  ▼ 1: Object
    _id : "University of Glasgow"
    totalFunding : 192500000
    avgFunding : 1458333.3333333333
    maxFunding : 3000000
    projectCount : 132
  ▼ 2: Object
    _id : "University College Cork"
    totalFunding : 121500000
    avgFunding : 1463855.4216867469
    maxFunding : 3000000
    projectCount : 83
  ▼ 3: Object
    _id : "TUS: Midlands Midwest"
    totalFunding : 177500000
    avgFunding : 1491596.638655462
    maxFunding : 3000000
    projectCount : 119
  ▼ 4: Object
    _id : "University of Cambridge"
    totalFunding : 162500000
    avgFunding : 1463963.963963964
    maxFunding : 3000000
    projectCount : 111
  ▼ 5: Object
    _id : "Cardiff University"
    totalFunding : 171500000
    avgFunding : 1478448.2758620689
    maxFunding : 3000000
    projectCount : 116
  ▼ 6: Object
    _id : "Trinity College Dublin"
    totalFunding : 185500000
    avgFunding : 1671171.1711711711
```

Discussion: This “mega-query” leverages \$facet to produce a dashboard-style breakdown for each university, integrating multiple metrics across four key profiles: funding, publications, subject breadth, and resource use. Each mini pipeline inside \$facet computes a different dimension, including ratios like average funding and resource diversity. The result is a composite research profile per institution — ideal for benchmarking, cluster analysis, or generating visual personas.

This type of query shows off the power of Compass as a visual analytics tool. It also reflects a high-level grasp of how MongoDB’s aggregation framework can be used to simulate multidimensional BI (business intelligence) reporting logic — merging statistical and structural views.

Compass/Atlas – Other Features Explored

In addition to querying and aggregation, Compass and Atlas offer several helpful features that were explored in this project. These functionalities are also supported and discussed in recent developer documentation and academic references, strengthening their relevance in real-world use:

- **Schema Tab:** Used to visually profile data distribution across fields and check for inconsistencies (e.g. string vs number types).
- **Explain Plan:** Helped analyze query execution performance and detect inefficient filters.
- **Validation Tab:** Reviewed for potential schema constraints and future schema validation setup.
- **Index Tab:** Explored to understand existing indexes and how they might optimize queries.
- **Dark Mode UI:** Activated for improved visibility during screencast capture.

These tools enhanced overall understanding of query structure, field consistency, and optimization opportunities. For example:

According to [MongoDB Atlas Docs](#), schema analysis and validation setup are crucial for enforcing best practices in data modeling.

In “Practical MongoDB Aggregations” by Paul Done, the author highlights the importance of Explain Plans and Index analysis for optimizing pipeline efficiency.

Articles such as '*MongoDB Aggregation Framework as an Analytical Engine*' (2021, IEEE Access) illustrate how these built-in Compass features mirror the needs of professional analysts in both academic and enterprise environments.

These references underscore the strategic importance of these tools for advanced data analytics workflows.

- A. [MongoDB Developer Hub](#): Offers articles, real-world tutorials, and case studies demonstrating MongoDB usage in data science and software engineering.
- B. "*Schema Design and Performance Optimization in MongoDB*" (IEEE International Conference on Big Data 2020): Emphasizes the use of schema exploration and indexing for optimizing aggregation.
- C. Discussions on [MongoDB Community Forums](#) provided valuable insights into Compass-based schema validation and visual aggregation.
- D. Blogs such as Studio 3T's MongoDB Best Practices gave practical guidance on using the Explain Plan and Schema tabs for query tuning.

Appendices

A. Youtube Links

Video 1: <https://youtu.be/ImHuT0ZtSoQ>

Video 2: <https://youtu.be/JtdJFTK6Gk8>

B. Links to Websites, Youtube clips, and other resources

Videos & Tutorials

- [MongoDB Crash Course – Traversy Media \(YouTube\)](#)
- [MongoDB Aggregation Framework Explained – Web Dev Simplified \(YouTube\)](#)
- [MongoDB Tutorial for Beginners – Programming with Mosh \(YouTube\)](#)
- [MongoDB Compass GUI Tutorial – Geeky Shows \(YouTube\)](#)

MongoDB University (Free Courses)

- [M001: MongoDB Basics](#)
- [M100: MongoDB for SQL Pros](#)
- [M121: The MongoDB Aggregation Framework](#)

Documentation & Guides

- [MongoDB Official Docs](#)
- [Getting Started Tutorial](#)
- [Introduction to MongoDB](#)
- [Databases and Collections](#)
- [CRUD Operations](#)
- [Query Documents](#)
- [Query Operators](#)
- [Regex Queries](#)
- [Aggregation Framework](#)
- [MongoDB Compass Docs](#)
- [MongoDB Atlas Docs](#)

Forums & Community

- [MongoDB Community Forums](#)
- [Stack Overflow – MongoDB](#)
- [Reddit – r/MongoDB](#)

LinkedIn Learning

- [NoSQL Essential Training](#)
- [MongoDB Courses & Atlas Tutorials](#)

C. Titles of journal articles, books or other publications used to complete the project

Books

1. **MongoDB: The Definitive Guide**

Authors: Shannon Bradshaw, Eoin Brazil, and Kristina Chodorow

Publisher: O'Reilly Media, 3rd Edition, 2019

Overview: A comprehensive guide covering MongoDB's core concepts, including querying, indexing, aggregation, transactions, and sharding. Suitable for both beginners and experienced developers.

2. **MongoDB in Action**

Authors: Kyle Banker, Peter Bakkum, Shaun Verch, Douglas Garrett, and Tim Hawkins

Publisher: Manning Publications, 2nd Edition, 2016

Overview: Offers practical examples and in-depth explanations of MongoDB's features, including replication, sharding, and deployment strategies.

3. **NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence**

Authors: Pramod J. Sadalage and Martin Fowler

Publisher: Addison-Wesley, 2012

Overview: Provides a concise introduction to NoSQL databases, discussing their types, use cases, and how they differ from traditional relational databases.

4. **Designing Data-Intensive Applications**

Author: Martin Kleppmann

Publisher: O'Reilly Media, 2017

Overview: Explores the principles and architectures behind scalable, reliable, and maintainable data systems, covering topics like data models, storage engines, and distributed systems.

Journal Articles

1. **A Comparison between Several NoSQL Databases with Comments and Notes**

Authors: B.G. Tudorica and C. Bucur

Conference: RoEduNet International Conference, 2011

Overview: Presents a comparative study of various NoSQL databases, analyzing their performance, scalability, and suitability for different applications.

2. **Survey on NoSQL Database**

Authors: J. Han, E. Haihong, G. Le, and J. Du

Conference: 6th International Conference on Pervasive Computing and Applications (ICPCA), IEEE, 2011

Overview: Provides a comprehensive survey of NoSQL databases, discussing their characteristics, data models, and classification based on the CAP theorem.

3. **NoSQL Database: New Era of Databases for Big Data Analytics – Classification, Characteristics and Comparison**

Authors: A.B.M. Moniruzzaman and S.A. Hossain

Journal: International Journal of Database Theory and Application, 2013

Overview: Discusses the emergence of NoSQL databases in the context of big data analytics, providing a classification and comparison of different NoSQL systems.